A Democratic Dividend in Trade? Evidence from a Flexible Empirical Implementation*

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Abstract: We study the causal effect of country-specific democratic regime change on bilateral trade, extending standard gravity empirics to 'heterogeneous gravity' estimated at the country-pair level. Our difference-in-differences implementation accounts for selection into regime change, multilateral resistance, globalisation effects and spatial dependence. We find average effects of around 40% higher exports for countries after thirty years in democracy, but demonstrate that these effects are driven by the democratic dividend for income per capita: the causal chain runs from democracy to economic prosperity to trade, and democracy appears to have a limited 'direct' effect on trade flows.

Keywords: trade gravity model, democratic regime change, monadic variables, heterogeneity, panel data, interactive fixed effects

JEL codes: F13, F14, P16, C23

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1 Introduction

Economists generally agree on the importance of good institutions for economic prosperity and the causal effect of becoming a democracy — a 'bundle' of institutions — on long-run income per capita is suggested to be on the order of 20-30% (Acemoglu et al. 2019, Boese-Schlosser & Eberhardt 2024).[1] Similarly, the relevance of trade for economic development is widely recognised among economists and policymakers alike: trade flows are associated with efficiency gains, technology transfer, and increased innovation activities (Costinot & Rodríguez-Clare 2014, Feyrer 2019). To the uninitiated, it may therefore come as a surprise that the literature on democratic institutions in their effect on bilateral trade flows is quite sparse. Furthermore, the effects of other important *country-specific* policies or characteristics (e.g. MFN tariffs, exchange rate arrangements, corporate taxation) on bilateral trade are also largely unexplored.

Counter-intuitively, the solution to this puzzle relates to the significant progress made in the literature: the structural gravity model[2] has been considerably strengthened by theoretical work deriving it from a wide class of microeconomic principles (Arkolakis et al. 2012, Costinot & Rodríguez-Clare 2014, Allen et al. 2020) and by contributions to the empirical methodology (Anderson & Van Wincoop 2003, Santos Silva & Tenreyro 2006, Baier & Bergstrand 2007).[3] The empirical path taken by the gravity literature to achieve credibility and rigour is also its demonstrable weakness, an empirical straitjacket of sorts: identification can be achieved by focusing on the narrow set of *dyadic* variables, with the analysis of country-specific (i.e. *monadic*) ones largely ignored due to adherence to the state-of-the-art of the structural gravity model. Exceptions include Heid et al. (2021) and Beverelli et al. (2024), who seek alternative means to identify monadic variables within the structural gravity model, involving intra-national sales, albeit for limited time series. The merit of this analysis hinges on the assumption that regime change affects domestic and international sales differentially and that quantifying this difference is economically meaningful, rather than establishing whether 'democracy causes higher trade' (like we do). Other work has sought to construct dyadic democracy variables to get around the identification problem (e.g. Yu 2010, Álvarez et al. 2018) but cannot speak to the impact of *national* democratic regime change. A substantial number of papers simply ignore the 'prescribed' set of fixed effects in their analysis of monadic variables (e.g. Dutt & Traca 2010, Francois & Manchin 2013). Finally, some studies (e.g. Eaton & Kortum 2002, Head & Ries

[1]These studies counter widespread scepticism among economists and political scientists about a positive 'democratic dividend'. It is suggested that allowing voters to remove an incumbent government through the power of the electoral process would drive up consumption and reduce the investment rate to the detriment of economic growth (e.g. Baum & Lake 2003, 334f). Studying China or the East Asian 'Tigers' suggests that democracy is not a necessary condition for development, but this form of cherry-picking of autocratic success stories ignores the established link between autocracy and the large variation in economic performance (Persson & Tabellini 2009, Knutsen 2012, Imam & Temple 2024) .

[2]For recent surveys see Head & Mayer (2014), Yotov et al. (2016), and Yotov (2024).

[3]These in effect prescribe what the gravity specification should look like, and how it should be estimated to address a range of econometric challenges, including the global network aspect of trade as well as concerns about the most appropriate estimator.

2008) argue that country-specific effects could be extracted in a second-step regression from the fixed effects of a structural gravity model. This approach has been undermined by the finding that in the Poisson Pseudo Maximum Likelihood (PPML) model favoured in the literature these fixed effects have a clear theoretical interpretation which rules out any role for monadic variables in a second-step analysis (Fally 2015, Beverelli et al. 2024).[4]

Within the political theory of trade, a 'direct democracy' model suggests that democratic regimes should act, under majority rule, in agreement with the trade preferences of the median voter (Mayer 1984, Dutt & Mitra 2002). Assuming that countries undergoing democratisation are labour-abundant, (capital-poor) workers gain from trade liberalisation in a standard Hecksher-Ohlin model.[5] Extension of the franchise implies that the median voter is a worker and therefore greater trade openness is adopted by the newly-elected government. A 'political support' or 'protection for sale' approach (Grossman & Helpman 1994, Rodrik 1995) would also allow existing interest groups (e.g. protectionist old-guard political and economic elites) to lobby for their preferred trade policies but their political influence is likely to be substantially eroded in the newly-democratised regime (Milner & Mukherjee 2009).[6] In addition, democratisation can have indirect economic effects by promoting legal and economic reforms which support all businesses, including trading firms (Rode & Gwartney 2012, Giuliano et al. 2013, Yue & Zhou 2018). The empirical literature on the effects of democratisation on trade is relatively scarce (and deviates from the 'prescribed' model specification of structural gravity) but, overall, finds a positive impact of 'more' democracy on the imports and exports of (capital-poor) developing countries (Milner & Kubota 2005, Eichengreen & Leblang 2008, Kono 2008, Tavares 2008, Aidt & Gassebner 2010, Yu 2010).

In this paper, we propose an estimation strategy to break the empirical straitjacket: we investigate the gravity relationship at the country-pair level, accounting for the manifestation of the trade network as well as regime change endogeneity.[7] Our implementation enables us to investigate the effect of *monadic* variables, such as democracy, on bilateral trade flows. We adopt a common factor framework to capture multilateral resistance and other forms of unobserved time-varying heterogeneity, treating the factors as nuisance terms in the same fashion as the standard pooled gravity model employs a myriad of fixed effects.[8] Following Boneva & Linton (2017) we use cross-section averages (CA)

[4]Freeman et al. (2021) recently suggested a two-step approach relying on the theoretical properties of Fally (2015) to calculate multilateral resistance control terms.

[5]In the context of global value chains (GVC) where exporting and importing activities are tightly linked, workers (and firms) in developing countries tend to gain from greater GVC participation (World Bank 2019), which itself both requires and encourages unilateral tariff liberalisation on final and intermediate goods (Baldwin 2010).

[6]In consolidated democratic regimes, there may be opportunities for new interest groups to emerge and 'buy protection'. This protectionism can be obfuscated by the implementation of trade policies less transparent than tariffs, such as quotas or product standards (Kono 2006).

[7]Democratic regime change is treated as endogenous given the link between economic integration and institutional change found in the literature (e.g. Acemoglu et al. 2005, Nunn & Trefler 2014, Puga & Trefler 2014).

[8]The common factor framework is popular in the cross-country empirical literature to capture total factor productivity, absorptive capacity and other unobserved heterogeneity (e.g. Eberhardt et al. 2013, Chirinko & Mallick 2017, De Visscher et al. 2020, Madsen et al. 2021). Our proposed implementation relates to existing work modelling

to proxy common factors (an approach pioneered by Pesaran 2006), in combination with the spirit of a heterogeneous treatment effects estimator by Chan & Kwok (2022):[9] we construct the CAs from 'economic mass' variables (GDP, population) in the samples of 'never-treated' exporters and importers, respectively, and test whether the 'information' they capture is equally relevant for the treated sample.[10] We estimate CA-augmented treatment equations for the bilateral trade flows of countries that experienced regime change during the sample period. A further correction addresses a practical problem in some country-pairs where the exporter or importer did experience regime change but no trade *ever* takes place between the countries.[11] Implementation is via PPML at the pair-level to address concerns over consistency and truncation. Our approach enables us to causally identify binary monadic variables since we capture time-varying unobservables with the CA-augmentation (with CAs varying at the *country-time* level for importers and exporters, respectively).[12] These factor proxies are constructed from data for countries which never democratise and are added to the treatment regression akin to a microeconometric control function approach (Olley & Pakes 1996, Levinsohn & Petrin 2003).[13]

The economic interpretation of our estimates crucially depends on our treatment of exporter and importer income per capita as additional covariates: (i) if we exclude these, our model identifies the 'total' effect of democracy on trade, but we cannot determine whether this is not the outcome of increased economic prosperity following regime change (the 'indirect effect' on trade); (ii) if, in line with structural gravity theory, we include these terms, our model can identify the 'direct effect' but our difference-in-differences estimates may be biased since these terms are 'bad controls' (controls which themselves are outcomes of the regime change treatment); fortunately, (iii) we can use the insights from gravity theory to suggest an alternative specification where the economic mass terms are imposed on the bilateral flow dependent variable and we can hence identify the 'direct effect' of regime change on trade without falling foul of the 'bad controls' trap.

We study the export flows of 84 countries which during 1950-2014 experienced 111 democratic

multilateral resistance in the gravity equation using common factors (e.g. Serlenga & Shin 2007, Chen et al. 2021), though these cannot capture the effect of monadic covariates.

[9] In this literature economists favour the use of binary indicators for democracy versus autocracy (Papaioannou & Siourounis 2008, Acemoglu et al. 2019, Boese-Schlosser & Eberhardt 2024), which we construct from V-Dem data.

[10] Pooled difference-in-differences estimators rely on the assumption of parallel trends before treatment. The Chan & Kwok (2022) estimator allows for non-parallel trends but requires the weaker assumption of equal average factor loadings on the common factors between treated and control samples — see Section 2 for details.

[11] We cannot estimate our pair-specific gravity regression in this case yet simply ignoring such pairs would induce selection bias. Our correction is once again based on cross-section averages, but for economic mass terms of those 'treated' exporters which had one or more trading relationships which never saw positive trade flows (we effectively weight these cross-country averages by the number of such no-trade pairs of each exporter).

[12] Like in a standard fixed effects model where the variable of interest is endogenous to time-invariant country effects, the solution to the identification problem is to include proxies for the unobserved heterogeneity: country fixed effects in the case of time-invariant heterogeneity, and multiple common factors with country pair-specific coefficients in the case of time-variant heterogeneity.

[13] Unlike in pooled difference-in-differences models, years for countries which are 'not yet' treated *never* enter the sample of control observations. This also bypasses concerns over negative weights in pooled difference-in-differences models (De Chaisemartin & d'Haultfœuille 2020).

regime changes and 55 reversals to autocracy.[14] Presenting our results in 5-year treatment intervals, we find that the 'total' effect of regime change on trade is very large: the average treatment effect after thirty years in democracy amounts to a 40% increase in exports, rising to 50% for countries which experienced just a single democratic regime change. Further, in line with recent results in the democracy-growth literature, export effects are lacklustre for countries with fewer than a decade and a half in democracy: insignificant and at times even negative-significant.[15] Our results show differences across *importer* regime types, whereby the variance of the export effect (confidence intervals) for *autocratic* destinations is substantially higher. However, once we re-specify our empirical models in line with structural gravity theory, these large long-run effects effectively disappear, whether we include income terms or account for these by adjusting the dependent variable: the causal chain is suggested to run from democracy to higher income to increased exports, with direct effects from democracy to exports quantitatively limited. The same patterns of results prevail if we study the effects of regime change on imports. We conduct a range of robustness checks including alternative cross-section average augmentations and alternative definitions of regime change, but our main findings remain qualitatively unchanged.

We make two important contributions to the literature: first, we introduce a simple but powerful empirical implementation of the gravity model of trade which allows for the identification of binary monadic variables. Second, we study the relationship between country-specific democratic regime change and bilateral trade flows, contributing to our understanding of the mechanisms and causal chain of effects underlying the sizeable democratic dividend for long-run economic prosperity.

2 Methodology

In this Section, we introduce our empirical estimator. We begin by detailing its five important elements and the literature(s) these are taken from. We then introduce the estimation equations, assumptions, implementation and inference. A more detailed exposition is provided in Appendix D. The final section details a specification test adapted from Chan & Kwok (2022).

2.1 A Multi-Faceted Implementation

Our difference-in-differences implementation to capture binary monadic variables in a factor-augmented heterogeneous gravity model has the following conceptual ingredients: (i) We build on the literature for heterogeneous parameter models (Swamy 1970, Pesaran & Smith 1995) and estimate the gravity model at the country pair-level. (ii) We can do so because we treat multilateral resistance

[14] These are all the countries which switched from autocracy to democracy at least once. An additional 53 countries remained autocratic throughout the sample period (these make up the control samples), while 28 countries were democratic throughout.

[15] These patterns can be rationalised by the upheaval following regime change and/or 'democratic overload' (see Gerring et al. 2005, Cervellati & Sunde 2014, Boese-Schlosser & Eberhardt 2024).

as unobserved time-varying heterogeneity and adopt a common factor framework to capture this in the model (Pesaran 2006, Bai 2009, Baier & Standaert 2024). (iii) We follow the gravity literature in estimating a generalised linear (PPML) model to address concerns over heteroskedastic errors and zero trade flows (Santos Silva & Tenreyro 2006), but introduce a variant estimated at the pair-level and embedded in a common factor framework using cross-section averages (CA) as factor proxies (Boneva & Linton 2017). (iv) Specifying democracy as a binary 'treatment' variable, we borrow from the treatment effects literature adopting common factors (Gobillon & Magnac 2016, Xu 2017, Chan & Kwok 2022) and estimate the gravity model at the pair-level for exporters i or destinations j which experienced regime change during the sample period (treated sample). We use the 'never-treated' exporters and importers (the control sample) to construct the factor proxies (cross-section averages) for causal identification of endogenous regime change, mimicking the Chan & Kwok (2022) strategy.[16] We further adapt their 'weak parallel trend test' to the three-way panel context to test the identifying assumption of expected factor-loading equality between treated and control samples. (v) Our pairwise regression is not identified for treated exporters or importers if their trade flows with *some* destinations remain zero in all time periods. We address this selection problem by including factor proxies from the zero-trade pairs in the treatment regressions. (vi) Finally, the inclusion or exclusion of exporter and destination per capita GDP (in logs, the 'economic mass' terms) poses econometric difficulties (if included) or concerns over the theory-consistency and hence interpretation of our empirical model (if excluded).

We refer to this as the heterogeneous PPML-CCE-DID estimator to highlight its most significant 'origins' (Santos Silva & Tenreyro 2006, Pesaran 2006, Chan & Kwok 2022, respectively).

2.2 Estimation and Inference

Using PPML, we estimate the following equation at the pair-level for exporters i and destinations j, either or both of which experienced democratic regime change:

$$\begin{aligned}
\mu_{ijt} \quad = \quad & exp[\alpha_{ij} + \gamma^1_{ij}\,\text{FTA}_{ijt} + \gamma^2_{ij}\,\text{Common Currency}_{ijt} + \theta_{ij} D_{it} + \eta_{ij} D_{jt} \qquad (1)\\
& \left\{ +\psi^i_{ij} ln(Y/L)_{it} + \psi^j_{ij} ln(Y/L)_{jt} \right\} \\
& +\delta^i_{ij}\,\overline{ln(Y)}^i_t + \kappa^i_{ij}\,\overline{ln(Pop)}^i_t + \delta^j_{ij}\,\overline{ln(Y)}^j_t + \kappa^j_{ij}\,\overline{ln(Pop)}^j_t \\
& +\delta^{i0}_{ij}\,\overline{ln(Y)}^{i0}_t + \kappa^{i0}_{ij}\,\overline{ln(Pop)}^{i0}_t + \delta^{j0}_{ij}\,\overline{ln(Y)}^{j0}_t + \kappa^{j0}_{ij}\,\overline{ln(Pop)}^{j0}_t + \epsilon_{ijt}] \qquad \forall ij,\, j \neq i.
\end{aligned}$$

[16]The intuition for the Chan & Kwok (2022) PCDID follows that of the control function approach in the analysis of production functions (Olley & Pakes 1996) using combinations of observed choice variables (e.g. materials) to construct a proxy of how firms react to changes in unobserved TFP.

The dependent variable represents the exports from i to j at time t. Our parameters of interest are θ_{ij} and η_{ij} which relate to the binary regime change indicators for exporters D_{it} and destinations D_{jt}.[17] They capture the individual treatment effects averaged over the treatment period (Chan & Kwok 2022, refer to these as ITET, individual treatment effects on the treated). FTA and Common Currency are dyadic controls. We ignore the second line of equation (1) for a moment and note that the third line represents the cross-section averages at time t ($\overline{ln(X)}_t^i = N^{-1} \sum_i ln(X)_{it}$) of the 'economic mass' variables employed in the gravity model (GDP, population; in logs). These are computed from respectively all exporters i (first two terms) and all destinations j (second two terms) which remained autocratic. The final line includes the error term and four more cross-section averages, computed from GDP and population data for exporters i which experienced regime change but for which trade flows to *some* destinations remained zero in all years and similarly for destinations j. In the construction of these cross-section averages we effectively weight the GDP and population data of exporters i (importers j) by the number of zero-trade partnerships they have.[18]

The second line of equation (1) refers to the exporter and destination income per capita terms: if we exclude these, we study the 'total' effect of regime change on trade but cannot determine whether democracy causes trade or trade increases because democracy causes higher income per capita. If we include them, we condition on income and hence identify the 'direct' effect of democracy on trade. Since in a DID framework their inclusion renders them as 'bad controls' (Angrist & Pischke 2008), we alternatively adjust the dependent variable accounting for the unit elasticity of these 'economic mass' terms: $\mu^*_{ijt} \equiv \mu_{ijt}/(Y_{jt}Y_{it})$.

Assumptions Since, by construction, the cross-section averages can have different effects/coefficients across country-pairs, the familiar 'parallel trend' assumption of standard pooled Difference-in-Differences models does not apply here — however, a weaker assumption of common expected factor loadings between treated and control samples is required for identification (see next section). The above setup can accommodate endogeneity of treatment D_{it} in terms of selection into treatment and the timing of treatment: D_{it} can be correlated, *inter alia*, with any parameters on the CA terms or with the additional controls (FTA, common currency). The main assumption we require for consistency of $\hat{\theta}_{ij}$ is that we capture all unobserved time-varying heterogeneity with our low-dimensional factor structure (Pesaran 2006, Bai 2009, Athey et al. 2021), hence, that ϵ_{ijt} is orthogonal to all elements of equation (1).[19] Remaining threats to identification arise from idiosyncratic country

[17]Following the literature (Pesaran 2006) we assume: $\theta_{ij} = \bar{\theta} + v_{ij}$ where $v_{ij} \sim IID(0, \Omega)$.

[18]Country A experiences democratic regime change and has 20 trade partners. For 15 of these we have non-zero trade flows for at least one year during the sample period (before or after regime change), but for 5 trade remained zero throughout. The CA terms with superscript $i0$ include five instances of Country A's GDP and population. Many countries like A will experience regime change but specific trade-pairs which never amount to non-zero flows and these cross-section averages will capture their unobserved time-varying heterogeneity with particular emphasis on those with comparatively many zero-trade partnerships.

[19]In robustness checks we consider alternative CA-augmentations with qualitatively identical results.

'shocks' which affect trade and may simultaneously trigger democratic regime change: for instance, a financial crisis or a natural resource discovery. Existing research suggests that financial crises have sizeable international dimensions (Cesa-Bianchi et al. 2019), while oil exploration is driven by global prices (Arellano et al. 2017) — both examples of common factors.

Post-Estimation and Inference We average treatment estimates $\hat{\theta}_{ij}$ to yield ATETs (see Appendix Table B-1) or relate them to treatment length (see Figure 1), using an M-estimator (Rousseeuw & Leroy 2005) to reduce the impact of outliers. These are Mean Group estimates: $\hat{\theta}^{MG} = N^{-1}\sum_i \omega_{ij}\hat{\theta}_{ij}$ (Pesaran & Smith 1995) with granular weights ω_{ij} and N the number of pairwise estimates. Our nonparametric variance estimator is $\widehat{\text{var}}(\hat{\theta}^{MG}) = [N(N-1)]^{-1}\sum_{i=1}^{N}(\hat{\theta}_{ij} - \hat{\theta}^{MG})^2$ following (Pesaran 2006).

2.3 Testing the weak parallel trend assumption

The reliability of standard difference-in-differences estimator lives and dies by the validity of the 'parallel trend' assumption: if the soon-to-be-treated unit was already on a different trajectory to the control unit(s) before treatment, then the standard difference-in-differences estimates are biased. In the Chan & Kwok (2022) PCDID case the estimator allows for non-parallel trends across panel units, most importantly between those in the treated and those in the control samples. Nevertheless, the estimator requires a weaker form of the parallel trend test for validity: the factor loadings on the common factors in the treatment equation on average have to be equal those in the control equation. In practice, this implies that the estimated parameters on the factor proxies in the treatment equation statistically should not differ from 1. The intuition of this test is to ask whether the information we extract from the control sample is equally relevant in the treatment sample: imagine a world in which democratic regime change only happens in rich countries, whereas poor countries always stay autocratic. The 'information' contained in the factor proxies constructed from the poor-country control sample is then likely to be quite uninformative about the unobservables driving outcomes in the rich-country treatment sample: countries at different ends of the income scale have different economic structures, financial systems, physical and human capital stocks, etc. — we would be proxying apples with oranges. Passing the Alpha test assures us that the unobservables are *on average the same* across treatment and control samples, that we are adopting apples to proxy for apples, or oranges for oranges.

In comparison to the Chan & Kwok (2022) Alpha test, in our setup using cross-section averages we cannot use their testing strategy, which employs the cross-section average of the residuals from an auxiliary regressions in the control sample. We devise an alternative Alpha test for the heterogeneous trade flow regressions augmented with cross-section averages by testing whether the averaged coefficients on the CA, $\bar{\hat{\delta}}^i$, $\bar{\hat{\kappa}}^i$ and $\bar{\hat{\delta}}^j$, $\bar{\hat{\kappa}}^j$ from equation (1), for exporters and destinations, respectively sum to 1. If the underlying null of equal average factor loadings between treatment and control samples is rejected, the treatment regression may be misspecified and hence deliver biased treatment estimates.

3 Data and Transformations

We use bilateral trade flows alongside exporter and importer GDP and Population from TRADHIST (Fouquin & Hugot 2016, version 4), combined with CEPII data for FTAs and currency unions.[20] For the export flow data we code the 'plausible zeros' as zero rather than missing. Our panel covers 1950-2014. Data on democracy is taken from the Varieties of Democracy (V-Dem) project (Coppedge et al. 2021): motivated by Mukand & Rodrik (2020) we primarily focus on the liberal democracy index but explore polyarchy and liberal component indices in extensions.[21] In a robustness check, we adopt the definition of democracy by Acemoglu et al. (2019). We discard country pairs with fewer than 14 observations — this is by the necessity of the demands of our heterogeneous panel estimator.

We adopt the full sample mean for liberal democracy over 1950-2014 (0.34) as the threshold for democratic regime change. The liberal democracy sample includes 822,xxx observations for 17,291 country pairs (average $T = 48$ years). Of these, 8,577 pairs are for regime changes in the exporter country, relating to 84 countries which trade with 15 (JAM) to 164 (TUR) destinations.[22] In robustness checks, we add/subtract a quarter or half a sample standard deviation to this mean index value to adopt a tighter/looser definition of regime change. For our baseline specification, we provide detailed distributions of regime change events and time spent in democracy in Appendix A. The latter suggest that the thousands of democracy estimates making up our results are relatively uniformly distributed across time spent in democracy.

4 Results

4.1 Main Results

Baseline Results: Export Effects Figure 1 presents the treatment effects by years spent in democracy (5-year intervals) and Appendix Table B-1 the equivalent ATETs (ignoring treatment length) alongside Pseudo-Alpha test results. We use an M-estimator (Rousseeuw & Leroy 2005) to compute outlier-robust means, and all specifications (unless discussed) satisfy the Pseudo-Alpha specification test. In each plot, we report robust mean estimates (diamonds), associated 95% confidence intervals (CI, coloured bars), and the number of country pairs for each mean estimate. All results presented are average treatment estimates for exporters, using the specification including

[20] All monetary values are in nominal British Pounds.

[21] These data are described in great detail in Boese (2019).

[22] There are a total of 169 destinations in this baseline model. For the analysis of regime change *in a destination*, Figure 1, panel (c), there are 8,434 pairs, relating to 84 countries.

dyadic controls.[23] We plot full sample results and distinguish by destination regime status (always autocratic or always democratic).[24]

Focusing on the full sample results (with blue CI) in Figure 1 panel (a) we can see volatile effects in the first 15 years, followed by treatment effects of .12, .2 and .4 (around 13%, 22%, and 49% higher exports) for countries which spent 16-20, 21-25 and 26-30 years in democracy, respectively.[25] The effect for countries which have been democratic for over 30 years is attenuated, just over .2 (22%). Mean estimates for the subsample of results where destinations are always democratic (teal-coloured CI) closely match the pattern described, whereas for the subsample of autocratic destinations (see-through CI) we observe much larger confidence intervals, which lead to mostly statistically insignificant results. Trade effects with autocratic destinations again show higher variation (larger CI).

These results indicate that democratic regime change *ultimately* causes an increase in export of 50-65% after 30 years in democracy. But these 'total effects' cannot speak to the causal chain of effects from democracy to trade, which we investigate next.

Theory-Consistent Specifications The theory-consistent structural gravity model includes 'economic mass' terms for exporter and importer countries. Econometrically, conditioning on respective per capita GDP of the trading partners speaks to the question whether democracy directly affects trade or whether this constitutes an indirect effect via economic development. We first discuss our results including 'economic mass' terms before addressing an econometric concern over 'bad controls'.

Panel (c) of Figure 1 reports the treatment effects by five-year intervals for all exporters which democratised. We can see that compared with the results in panel (a) for all treatment length intervals the overall effects are substantially attenuated: treatment effects for all country pairs are mostly statistically insignificant and of modest magnitude if significantly different from zero (e.g. exporters with more than 30 years of democratic experience). With minor deviations, this result holds for the results in panel (d) where we only look at countries which experienced democratic regime change once (and never reverted to autocracy). Conditioning on economic development, we can suggest

[23] Our gravity equation includes dyadic indicators for free trade agreements (FTAs) and common currency arrangements. These are prime objects of analysis in the current pooled structural gravity model paradigm. In our results the mean coefficient on entering into an FTA has a very reasonable magnitude: around 0.15 ($t = 6.23$, $N = 1,346$) if we compute an outlier-robust mean in the sample of democratising exporters, suggesting 16% higher exports. The robust mean for common currency unions is -0.14 ($t = -1.32$, $N = 220$) among democratising exporters — most of these estimates (note the very small sample size) are for regional African monetary unions. Excluding these dyadic controls for FTAs and common currency unions yields qualitatively identical results.

[24] We exclude estimates for destinations which themselves experienced regime change for the conceptual reason that these conflate the effects when destinations are autocracies and democracies in a non-trivial manner and for the practical reason that their variability hampers ease of illustration.

[25] As is found in the existing literature on democracy and growth (Acemoglu et al. 2019, Boese-Schlosser & Eberhardt 2023), the initial period after regime change may be accompanied by economic 'hardship' or stagnation. Other than this 'tumultuous youth' experienced by many democracies (Gerring et al. 2005), our robustness checks adopting alternative democracy thresholds suggest that this finding at least in parts is due to the way we define regime change.

that the direct effect of democratic regime change is negligible and that the causal chain runs from democracy to higher income per capita to higher trade.

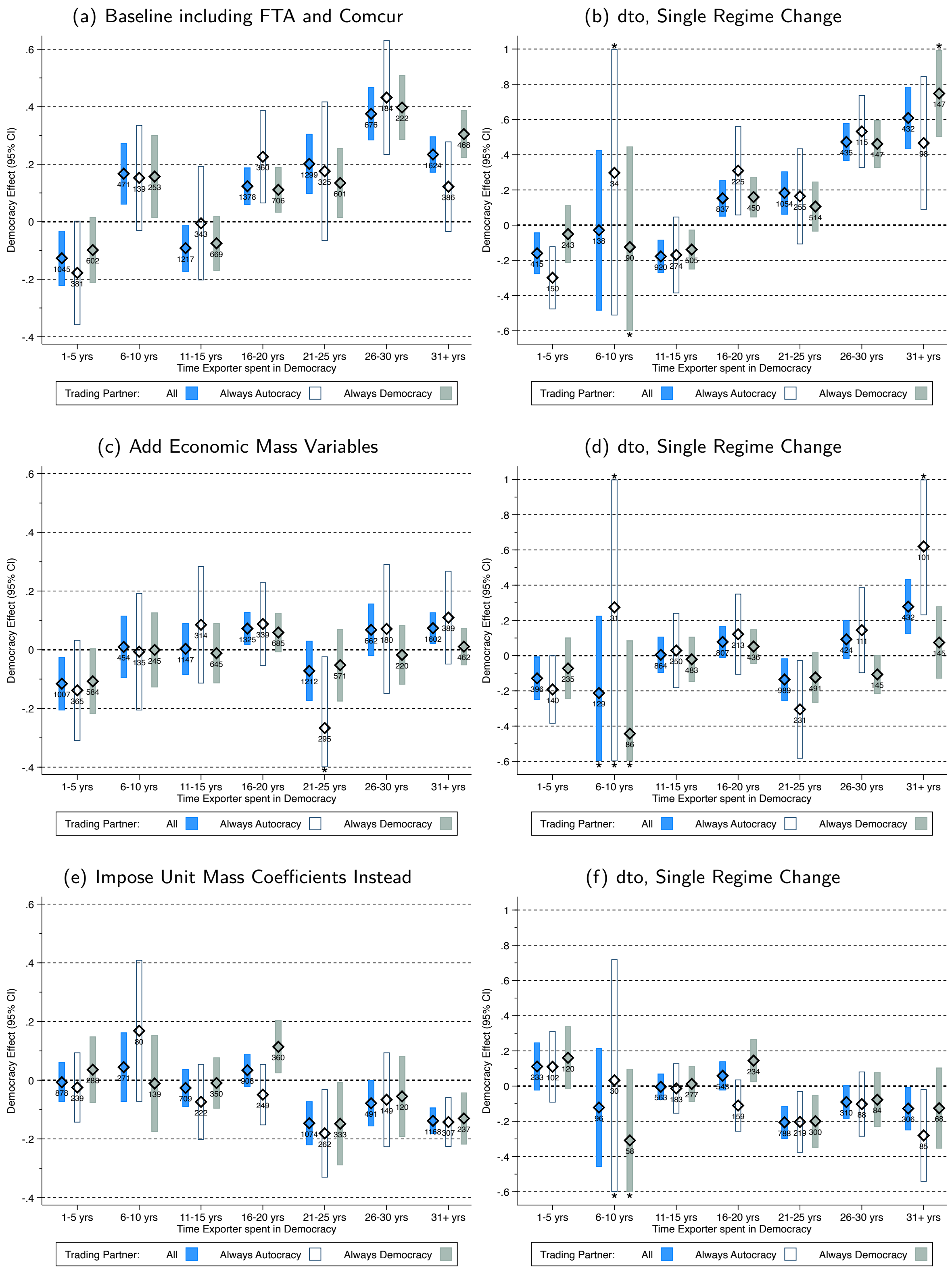
However, given our difference-in-differences specification, we recognise that since we *know* income per capita to be causally affected by democratic regime change (Acemoglu et al. 2019, Eberhardt 2021, Boese-Schlosser & Eberhardt 2024) these economic mass terms constitute 'bad controls' (Angrist & Pischke 2008). Fortunately, theory can help us avoid this problem given that the economic mass (GDP) terms in a structural gravity model are theoretically predicted to have unit elasticities (Head & Mayer 2014, see discussion in). We proceed to adjust our trade flow dependent variable by accounting for economic mass, $FLOW_{ij}/(GDP_i * GDP_j)$, and drop the income terms from the estimation equation. Panels (e) and (f) present the full results and those for countries with just a single regime change event for this specification. In line with the conclusion for the theory-consistent specifications in panels (c) and (d) we find severely attenuated and largely statistically insignificant treatment effects of regime change on trade flows. It should be noted that the cross-section averages capturing the unobserved time-varying heterogeneity in *destination* countries which never democratised fail the Alpha test in this specification ($p = .019$) whereas those for *exporter* countries do not ($p = .601$). Since our focus is here on the export effects of regime change, the overall treatment effects should not be affected by this misspecification.

Our findings thus support the argument of a dominant causal chain from democracy to economic prosperity to trade, whereas direct trade effects of democratic regime change appear to be small.[26]

Import Effects Our above analysis has asked about the effect of exporter regime change on their exports. Our model further allow us to quantify how export volume of exporters is affected if a destination country experiences regime change; an alternative interpretation is simply that of the import effects of regime change. In Figure 2 we present result plots for the same specifications as in our previous analysis. Full sample for *all exporters* are presented with CI in marigold, the subsample results are now for *exporters* which are always democratic (CI in pink) or always autocratic (CI in grey). Baseline results in panels (a) and (b) are qualitatively very similar to the export results in Figure 1, including for destination sub-samples, albeit with the absence of a negative effect for countries with 11-15 years in democracy and frequently larger magnitudes. Crucially, the results speaking to theory-consistent gravity estimation in panels (c) to (f) again report substantially attenuated treatment effects. It bears reminding that the final results in panels (e) and (f) are based on the model with economic mass imposed which fails the Alpha specification test, hence we cannot be as confident as in the export analysis that the models with economic mass variables in (c) and (d) are not misspecified.

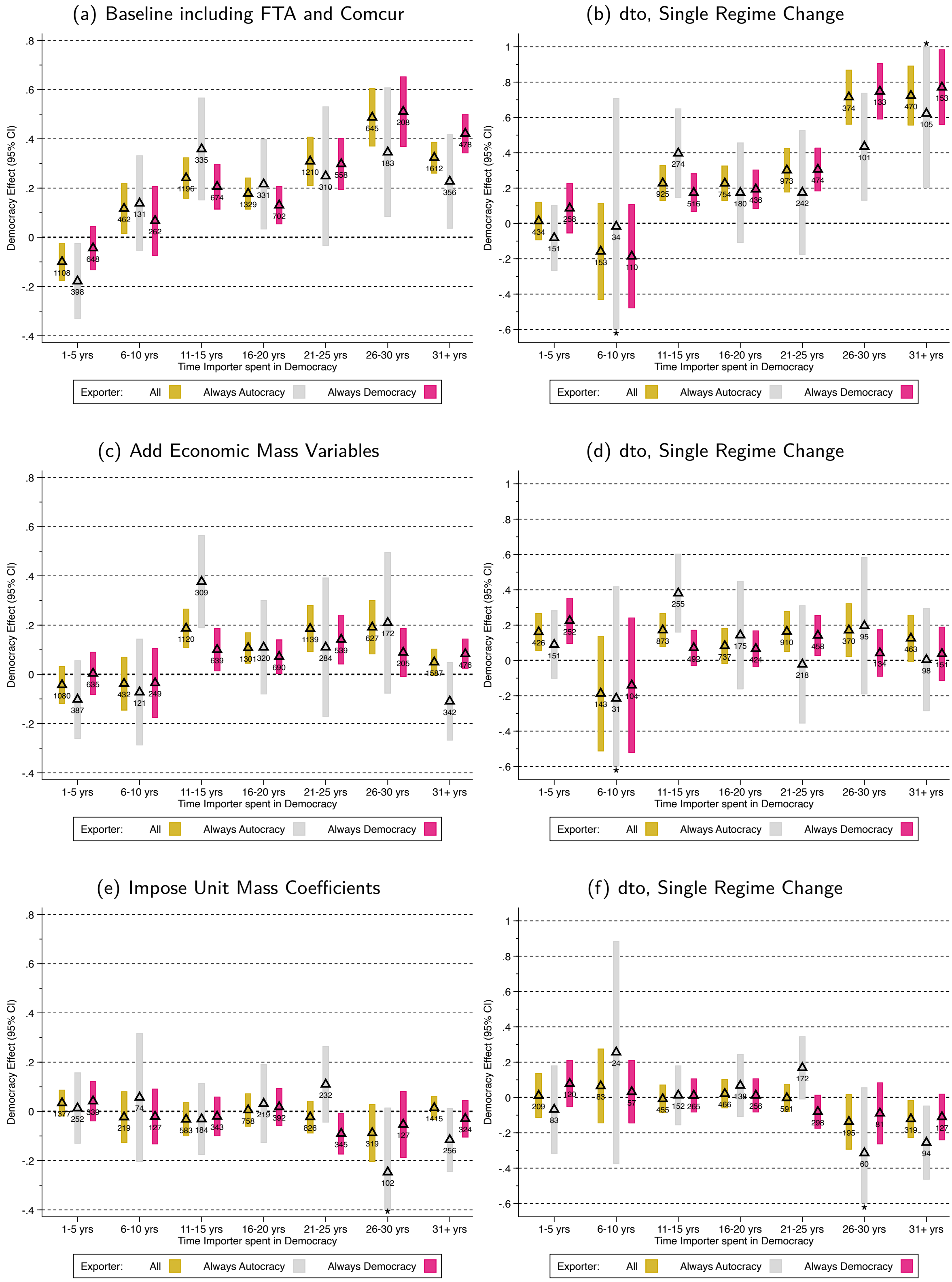
[26]In the specifications with income, outlier-robust mean estimates for FTA are around 0.11 ($t = x.xx$, $N = x, x$) in the sample of democratising exporters, and around 0.21 ($t = x.xx$, $N = x, x$) in the specification with economic mass imposed with unit coefficient, suggesting 12-23% higher exports.

Figure 1: Exporter Democratic Regime Change and Trade Flows — PPML-CCE-DID Results



Notes: These plots present ATET (using robust mean estimates) by time spent in democracy using hollow diamond markers. The bars show the 95% confidence interval (* indicates we curtailed these to aid presentation) and we report the underlying number of pairwise estimates. These are not event plots: e.g. exporters with 21 years in democracy only appear in the ATET for 21-25 years. A value of 0.2 suggests that the exporter's switch from autocracy to democracy caused a 20% increase in exports. Median length in liberal democracy for all exporter countries is 20 years. Results are also disaggregated by destination regime status (autocracy, democracy). Panels (a) and (b) are based on the benchmark model including FTA and ComCur dummies — results are virtually identical if we exclude the latter two. In panels (c) and (d) the specifications additionally include exporter and destination per capita GDP (in logs). In panels (e) and (f) we instead impose unit mass on the trade flow variable. Panels (b), (d) and (f) restrict the sample to exporters which shifted from autocracy to democracy once (and did not revert).

Figure 2: Destination Democratic Regime Change and Trade Flows — PPML-CCE-DID Results



Notes: These plots present ATET (using robust mean estimates) by time spent in democracy using hollow triangle markers. The bars show the 95% confidence interval and the underlying number of pairwise estimates is indicated. These plots are the equivalent of those in Figure 1 but from the importer's perspective (destination) and provide insight into the causal effect of democratic regime change on *imports*. Sub-sample results are therefore by exporter regime.

4.2 Robustness Checks and Variations

Alternative CA specifications Our results are based on empirical specifications incorporating cross-section averages (CAs) of GDP and Population for exporters and for destinations — the second row of equation (1). In Appendix Figure C-1 we adopt alternative models (i) limiting CAs to those for GDP only in panels (a) and (b), and (ii) adding CAs for country total exports to GDP and population in panels (c) and (d).[27] Result patterns are qualitatively very similar to the baseline estimates in panel (a) of Figure 1. Note that the specification with income and extended CA in panel (d) fails the Alpha test.

Ignoring Zero Trade Flows One feature of our implementation is the adjustment for treated countries which had zero export flows with specific destinations throughout the entire sample period. In Panels (e) and (f) of Appendix Figure C-1 we illustrate what happens to our results if we ignore this selection problem (excluding the null effect of democratic regime change) by omitting the final row of terms in equation (1) (except the error term). For the baseline model, patterns match those of our earlier results, but magnitudes are typically inflated. However, when we include income terms the treatment effects are once again substantially attenuated, supporting our previous findings.

Alternative Definitions of Democracy All of the above results are based on taking an arbitrary (mean) cut-off of the liberal democracy index to dichotomise regimes. As a robustness check, we still use the 1950-2014 V-Dem index mean but add or subtract one-quarter or one-half of its standard deviation to define regime change. Results in Appendix Figure C-2 panels (c), (e), (g) and (i) present baseline results for more liberal or more conservative definitions of democracy. Across all specifications, trade effects are (eventually) rising with years in democracy. The accompanying panels (d), (f), (h), and (j) present results for specifications including the income variables. Across all specifications, more prominently in the more liberal definitions of democracy, trade effects are substantially attenuated and frequently insignificantly different from zero. Note that the models in panels (g) and (h) (Mean $+1/4$ SD) fail the Alpha specification test.

We also adopt the categorisation of Acemoglu et al. (2019) which is based on Polity IV and Freedom House definitions, in addition to detailed case analysis (following Papaioannou & Siourounis 2008). Results in panel (k) and (l) of Appendix Figure C-2 again share the now familiar patterns for the baseline specification and the model with income terms.

[27] These setups are motivated by (i) the CCE rank condition, the assumption that the span of the CAs encompasses the unobserved factors (Boneva & Linton 2017), and (ii) the insight that including 'too many' CAs can be harmful (Juodis 2022).

4.3 Extensions

Average Trajectory of Treatment Effects All our results relate the treatment effect to the time spent in democracy. The estimates for a country with, say, 23 years in democracy are *only* contained in the robust mean for '21-25 years' — we cannot make any statements about the trajectory of treatment effects for said country from the early years in democracy to the 23 years they eventually accumulated. In panel (a) of Appendix Figure C-3 we follow the strategy in Boese-Schlosser & Eberhardt (2024) and add dummies for the first fifteen years in democracy to our gravity regressions and present outlier-robust means for each of these first 15 years in democracy as well as for all years over 15 ('Long Run'). This mimics an event study analysis, at least for the initial periods, in the context of our heterogeneous gravity model. The export effects of regime change rise to over 20% in the first dozen years and then increase further to around 40% by 14 years in democracy. The average country with 16-62 years in democracy enjoys 22% higher exports. These results suggest that our earlier finding of an inverted U-pattern in the first 15 years is not representative of the *trajectory* of the average democratiser, but reflects the characteristics of the countries *with only a few years in democracy*. Note that the two findings are not contradictory, they simply measure different things. However, these estimates are again substantially attenuated in panel (b) where we add the income terms.

Building Blocks of Liberal Democracy Political scientists commonly adopt minimal definitions of democracy which build on Dahl's (1971) work and emphasise democracy as a "competitive struggle for the people's vote" (Schumpeter 1942, 26). They argue that the accountability of a regime can only come from the power of the electorate to extend or withdraw the mandate of the leader(s): "Democracy is a system in which parties lose elections" (Przeworski 1991, 10). Economists commonly put greater emphasis on the 'Northian' institutions (North 1981, North & Weingast 1989), a suite of civil rights (the 'rule of law') and constraints on the executive. These institutional building blocks ensure that the government is providing the right incentives and opportunities to private businesses, which entails the reduction of market frictions (e.g. financial development) and the facilitation of transactions more generally, most notably foreign trade (Besley 1995).

In our empirical analysis above we adopt liberal democracy which combines both these definitions. The building blocks of liberal democracy are the V-Dem proxies for 'electoral democracy' and the 'liberal component' — see Boese-Schlosser & Eberhardt (2023). We dichotomise these indices at their full sample means to construct treatment dummies for 'regime change'. Our results in panels (c) and (e) of Appendix Figure C-4 reveal that these building blocks have comparatively similar effect patterns on trade flows when we ignore theory-consistent modelling and have again substantially attenuated effects if we include income terms in panels (d) and (f).

5 Concluding Remarks

In this paper, we introduced an empirical approach which seeks to satisfy the rigour of the structural gravity model and enable the analysis of country-specific variables within a 'heterogeneous' gravity model. This opens up the opportunity to study a vast array of trade determinants which the current literature had to ignore in its adherence to the empirical straitjacket of the pooled gravity model. We employ this new machinery to investigate one of the most pertinent questions in development economics, namely that of the impact of institutional change on economic prosperity: does democracy cause greater trade flows? Our first set of results for our sample from 1950 to 2014 indicate that the average country with three decades of democratic experience can expect high economic benefits in the form of a 40% increase in exports. This effect is driven by increased exports to other democracies, whereas the magnitude of change in exports to autocracies is a lot more varied.

However, this finding only captures the end of a causal chain, since our subsequent gravity theory-consistent specifications reveal that democratic regime change does not have a direct causal effect on trade flows: our results suggest that instead democracy causes higher economic prosperity, which in turn causes increased trade.

This is just a first attempt at highlighting the heterogeneous causal effects of institutional change in the context of the empirical gravity model. Given the rich literature in political economy and political science, we expect this methodology to offer important insights in future research. The wider implications of our heterogeneous gravity model, allowing for the study of time-varying binary monadic variables, has significant potential to inform policy on a range of trade determinants which previously had to be ignored.

References

Acemoglu, D., Johnson, S. & Robinson, J. A. (2005), 'The rise of europe: Atlantic trade, institutional change, and economic growth', American Economic Review **95**(3), 546–579.

Acemoglu, D., Naidu, S., Restrepo, P. & Robinson, J. A. (2019), 'Democracy does cause growth', Journal of Political Economy **127**(1), 47–100.

Aidt, T. S. & Gassebner, M. (2010), 'Do autocratic states trade less?', The World Bank Economic Review **24**(1), 38–76.

Allen, T., Arkolakis, C. & Takahashi, Y. (2020), 'Universal gravity', Journal of Political Economy **128**(2), 393–433.

Álvarez, I. C., Barbero, J., Rodríguez-Pose, A. & Zofío, J. L. (2018), 'Does institutional quality matter for trade? institutional conditions in a sectoral trade framework', World Development **103**, 72–87.

Anderson, J. E. & Van Wincoop, E. (2003), 'Gravity with gravitas: a solution to the border puzzle', American Economic Review **93**(1), 170–192.

Angrist, J. D. & Pischke, J.-S. (2008), Mostly harmless econometrics, Princeton University Press.

Arellano, C., Bai, Y. & Lizarazo, S. V. (2017), Sovereign risk contagion, NBER Working Paper Series 24031, National Bureau of Economic Research.

Arkolakis, C., Costinot, A. & Rodríguez-Clare, A. (2012), 'New trade models, same old gains?', American Economic Review **102**(1), 94–130.

Athey, S., Bayati, M., Doudchenko, N., Imbens, G. & Khosravi, K. (2021), 'Matrix completion methods for causal panel data models', Journal of the American Statistical Association **116**(536), 1716–1730.

Bai, J. (2009), 'Panel data models with interactive fixed effects', Econometrica **77**(4), 1229–1279.

Baier, S. L. & Bergstrand, J. H. (2007), 'Do free trade agreements actually increase members' international trade?', Journal of International Economics **71**(1), 72–95.

Baier, S. & Standaert, S. (2024), 'Gravity, globalization and time-varying heterogeneity', European Economic Review **163**, 104679.

Baldwin, R. (2010), 'Unilateral tariff liberalisation', The International Economy **2010**(14), 10–43.

Baum, M. A. & Lake, D. A. (2003), 'The political economy of growth: democracy and human capital', American Journal of Political Science **47**(2), 333–347.

Besley, T. (1995), 'Property rights and investment incentives: Theory and evidence from ghana', Journal of Political Economy **103**(5), 903–937.

Beverelli, C., Keck, A., Larch, M. & Yotov, Y. V. (2024), 'Institutions, trade, and development: identifying the impact of country-specific characteristics on international trade', Oxford Economic Papers **76**(2), 469–494.

Boese-Schlosser, V. A. & Eberhardt, M. (2023), How does democracy cause growth?, Discussion Paper SP-V-2023-501, Berlin: WZB.

Boese-Schlosser, V. A. & Eberhardt, M. (2024), 'Democracy doesn't always happen overnight: regime change in stages and economic growth', Review of Economics and Statistics (forthcoming).

Boese, V. A. (2019), 'How (not) to measure democracy', International Area Studies Review **22**(2), 95–127.

Boneva, L. & Linton, O. (2017), 'A discrete-choice model for large heterogeneous panels with interactive fixed effects with an application to the determinants of corporate bond issuance', Journal of Applied Econometrics **32**(7), 1226–1243.

Cervellati, M. & Sunde, U. (2014), 'Civil conflict, democratization, and growth: Violent democratization as critical juncture', Scandinavian Journal of Economics **116**(2), 482–505.

Cesa-Bianchi, A., Martin, F. E. & Thwaites, G. (2019), 'Foreign booms, domestic busts: The global dimension of banking crises', Journal of Financial Intermediation **37**, 58–74.

Chan, M. K. & Kwok, S. S. (2022), 'The PCDID approach: difference-in-differences when trends are potentially unparallel and stochastic', Journal of Business & Economic Statistics **40**(3), 1216–1233.

Chen, M., Fernández-Val, I. & Weidner, M. (2021), 'Nonlinear factor models for network and panel data', Journal of Econometrics **220**(2), 296–324.

Chirinko, R. S. & Mallick, D. (2017), 'The substitution elasticity, factor shares, and the low-frequency panel model', American Economic Journal: Macroeconomics **9**(4), 225–253.

Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., Bernhard, M., Cornell, A., Fish, M. S., Gastaldi, L. et al. (2021), V-Dem Dataset v11, Varieties of Democracy Institute, Available at: https://doi.org/10.23696/vdemds21.

Costinot, A. & Rodríguez-Clare, A. (2014), Trade theory with numbers: Quantifying the consequences of globalization, in 'Handbook of International Economics', Vol. 4, Elsevier, pp. 197–261.

Dahl, R. A. (1971), Polyarchy: Participation and opposition, Yale University Press, New Haven and London.

De Chaisemartin, C. & d'Haultfœuille, X. (2020), 'Two-way fixed effects estimators with heterogeneous treatment effects', American Economic Review **110**(9), 2964–96.

De Visscher, S., Eberhardt, M. & Everaert, G. (2020), 'Estimating and testing the multicountry endogenous growth model', Journal of International Economics **125**, 103325.

Dutt, P. & Mitra, D. (2002), 'Endogenous trade policy through majority voting: an empirical investigation', Journal of International Economics **58**(1), 107–133.

Dutt, P. & Traca, D. (2010), 'Corruption and bilateral trade flows: extortion or evasion?', Review of Economics and Statistics **92**(4), 843–860.

Eaton, J. & Kortum, S. (2002), 'Technology, geography, and trade', Econometrica **70**(5), 1741–1779.

Eberhardt, M. (2021), Democracy, growth, heterogeneity, and robustness, CEPR Discussion Paper 16719, CEPR.

Eberhardt, M., Helmers, C. & Strauss, H. (2013), 'Do spillovers matter when estimating private returns to R&D?', Review of Economics and Statistics **95**(2), 436–448.

Eichengreen, B. & Leblang, D. (2008), 'Democracy and globalization', Economics & Politics **20**(3), 289–334.

Fally, T. (2015), 'Structural gravity and fixed effects', Journal of International Economics **97**(1), 76–85.

Feyrer, J. (2019), 'Trade and income—exploiting time series in geography', American Economic Journal: Applied Economics **11**(4), 1–35.

Fouquin, M. & Hugot, J. (2016), Two centuries of bilateral trade and gravity data: 1827-2014, CEPII Working Paper 2016-14.

Francois, J. & Manchin, M. (2013), 'Institutions, infrastructure, and trade', World Development **46**, 165–175.

Freeman, R., Larch, M., Theodorakopoulos, A. & Yotov, Y. V. (2021), 'Unlocking new methods to estimate country-specific trade costs and trade elasticities', CESifo Working Paper No. 9432 .

Gerring, J., Bond, P., Barndt, W. T. & Moreno, C. (2005), 'Democracy and economic growth: A historical perspective', World Politics **57**(3), 323–364.

Giuliano, P., Mishra, P. & Spilimbergo, A. (2013), 'Democracy and reforms: evidence from a new dataset', American Economic Journal: Macroeconomics **5**(4), 179–204.

Gobillon, L. & Magnac, T. (2016), 'Regional policy evaluation: Interactive fixed effects & synthetic controls', Review of Economics and Statistics **98**(3), 535–51.

Grossman, G. M. & Helpman, E. (1994), 'Protection for sale', American Economic Review **84**(4), 833–850.

Head, K. & Mayer, T. (2014), Gravity equations: Workhorse, toolkit, and cookbook, in 'Handbook of International Economics', Vol. 4, Elsevier, pp. 131–195.

Head, K. & Ries, J. (2008), 'FDI as an outcome of the market for corporate control: Theory and evidence', Journal of International Economics **74**(1), 2–20.

Heid, B., Larch, M. & Yotov, Y. V. (2021), 'Estimating the effects of non-discriminatory trade policies within structural gravity models', Canadian Journal of Economics **54**(1), 376–409.

Huntington, S. P. (1991), 'Democracy's third wave', Journal of Democracy **2**(2), 12–34.

Imam, P. A. & Temple, J. R. (2024), 'Political institutions and output collapses', European Journal of Political Economy **85**, 102573.

Juodis, A. (2022), 'A regularization approach to common correlated effects estimation', Journal of Applied Econometrics **37**(4), 788–810.

Knutsen, C. H. (2012), 'Democracy and economic growth: A survey of arguments and results', International Area Studies Review **15**(4), 393–415.

Kono, D. Y. (2006), 'Optimal obfuscation: Democracy and trade policy transparency', American Political Science Review **100**(3), 369–384.

Kono, D. Y. (2008), 'Democracy and trade discrimination', The Journal of Politics **70**(4), 942–955.

Levinsohn, J. & Petrin, A. (2003), 'Estimating production functions using inputs to control for unobservables', Review of Economic Studies **70**(2), 317–341.

Madsen, J. B., Minniti, A. & Venturini, F. (2021), 'Wealth inequality in the long run: A schumpeterian growth perspective', The Economic Journal **131**(633), 476–497.

Mayer, W. (1984), 'Endogenous tariff formation', The American Economic Review **74**(5), 970–985.

Milner, H. V. & Kubota, K. (2005), 'Why the move to free trade? democracy and trade policy in the developing countries', International Organization **59**(1), 107–143.

Milner, H. V. & Mukherjee, B. (2009), 'Democratization and economic globalization', Annual Review of Political Science **12**(1), 163–181.

Mukand, S. W. & Rodrik, D. (2020), 'The political economy of liberal democracy', Economic Journal **130**(627), 765–792.

North, D. C. (1981), Structure and change in economic history, New York: Norton.

North, D. C. & Weingast, B. R. (1989), 'Constitutions and commitment: the evolution of institutions governing public choice in seventeenth-century england', Journal of Economic History **49**(4), 803–832.

Nunn, N. & Trefler, D. (2014), 'Domestic institutions as a source of comparative advantage', Handbook of International Economics **4**, 263–315.

Olley, G. S. & Pakes, A. (1996), 'The dynamics of productivity in the telecommunications equipment industry', Econometrica **64**(6), 1263–1297.

Papaioannou, E. & Siourounis, G. (2008), 'Democratisation and growth', Economic Journal **118**(532), 1520–1551.

Persson, T. & Tabellini, G. (2009), 'Democratic capital: The nexus of political and economic change', American Economic Journal: Macroeconomics **1**(2), 88–126.

Pesaran, M. H. (2006), 'Estimation and inference in large heterogeneous panels with a multifactor error structure', Econometrica **74**(4), 967–1012.

Pesaran, M. H. & Smith, R. (1995), 'Estimating long-run relationships from dynamic heterogeneous panels', Journal of Econometrics **68**(1), 79–113.

Przeworski, A. (1991), Democracy and the market: Political and economic reforms in Eastern Europe and Latin America, Cambridge University Press.

Puga, D. & Trefler, D. (2014), 'International trade and institutional change: Medieval venice's response to globalization', Quarterly Journal of Economics **129**(2), 753–821.

Rode, M. & Gwartney, J. D. (2012), 'Does democratization facilitate economic liberalization?', European Journal of Political Economy **28**(4), 607–619.

Rodrik, D. (1995), Political economy of trade policy, in G. G. & K. Rogoff, eds, 'Handbook of International Economics', Vol. 3, Elsevier, chapter 28, pp. 1457–1494.

Rousseeuw, P. J. & Leroy, A. M. (2005), Robust regression and outlier detection, John Wiley & Sons.

Santos Silva, J. & Tenreyro, S. (2006), 'The log of gravity', Review of Economics and Statistics **88**(4), 641–658.

Schumpeter, J. A. (1942), Capitalism, socialism and democracy, Harper & Brothers: London and New York.

Serlenga, L. & Shin, Y. (2007), 'Gravity models of intra-eu trade: application of the ccep-ht estimation in heterogeneous panels with unobserved common time-specific factors', Journal of Applied Econometrics **22**(2), 361–381.

Swamy, P. A. (1970), 'Efficient inference in a random coefficient regression model', Econometrica pp. 311–323.

Tavares, J. (2008), 'Trade, factor proportions, and political rights', The Review of Economics and Statistics **90**(1), 163–168.

World Bank (2019), World development report 2020: Trading for development in the age of global value chains, The World Bank.

Xu, Y. (2017), 'Generalized synthetic control method: Causal inference with interactive fixed effects models', Political Analysis **25**(1), 57–76.

Yotov, Y. V. (2024), 'The evolution of structural gravity: The workhorse model of trade', Contemporary Economic Policy **42**(4), 578–603.

Yotov, Y. V., Piermartini, R., Larch, M. et al. (2016), An advanced guide to trade policy analysis: The structural gravity model, Geneva: World Trade Organization.

Yu, M. (2010), 'Trade, democracy, and the gravity equation', Journal of Development Economics **91**(2), 289–300.

Yue, J. & Zhou, S. (2018), 'Democracy's comparative advantage: Evidence from aggregated trade data, 1962–2010', World Development **111**, 27–40.

Online Appendix – Not Intended for Publication

A Data, Sample Makeup, and Descriptives

A.1 Sample makeup

The baseline specification (liberal democracy) has 8,xxx estimates for regime change. In Appendix Table B-1 panel column (1) the robust mean ATET of 0.xxx is computed using an M-estimator which weights each estimate (0 to 1). In our graphical presentation in the main text, those estimates with zero weight are not counted/included, our total here is 7,xxx pairwise estimates for xx exporters (and up to xxx destinations).

61 countries (73%) had a single regime change from democracy to autocracy, 12 had two, 6 three and 2 four — 3 countries only reverted from autocracy to democracy. The mean (median) time spent in democracy is 23 (22) years. Appendix Table A-1 provides further details.

A.2 Descriptives

Panel (a) of Appendix Figure A-1 provides the distribution of regime change events (111 democratisations, 55 autocratisations) in the exporter countries of our sample. We distinguish the Cold War and Post-Cold War periods and highlight the 'Third Wave of Democratisation' (Huntington 1991) in the years following 1989 (33 democratic regime change events during 1990-95).

Panel (b) indicates the distribution of years spent in democracy (using the dyadic data) for four different samples, defined by the destination: (i) all pairs, (ii) pairs where the importer is a democracy throughout the sample period, (iii) pairs where the importer experiences regime change during the sample period, and (iv) pairs where the importer is autocratic throughout. In this and all results plots by construction the country experiencing regime change is always the exporter i, and on the x-axis we can read off the total number of years the exporter country has spent in democracy. With exception of the democratiser sub-sample,[28] these distributions suggest that the thousands of democracy estimates making up the results we present below are relatively uniformly distributed across these different periods in democracy,[29].

[28] Here the youngest democracies only make up 7% of the sample, 24% of estimates are for countries with 11-20 years of experience, and 35% each for 21-30 and over 30 years, respectively.

[29] Around one-quarter of estimates are for exporters with up to 10 years experience of democracy, 30-35% for those with 11 to 20 years, 24-28% for those with 21 to 30 years and 13-20% for those with more than 30 years in democracy.

Table A-1: Sample Makeup: Exporters

	ISO	Exporter	Start	Pairs	Demo	Reversal	Years
1	ALB	Albania	1950	54	1	0	23
2	ARG	Argentina	1950	124	4	3	38
3	ARM	Armenia	1992	43	0	1	3
4	BEN	Benin	1959	64	1	0	24
5	BFA	Burkina Faso	1954	81	1	0	16
6	BGR	Bulgaria	1950	102	1	0	25
7	BIH	Bosnia & Herzegovina	1994	53	1	0	18
8	BLR	Belarus	1992	111	0	1	5
9	BOL	Bolivia	1950	64	1	0	30
10	BRA	Brazil	1950	132	1	0	29
11	BTN	Bhutan	1980	34	1	0	7
12	CHL	Chile	1950	95	2	2	46
13	CIV	Cote d'Ivoire	1960	134	1	0	13
14	COG	Republic of Congo	1950	62	2	2	2
15	COL	Colombia	1950	97	2	1	35
16	COM	Comores	1980	44	1	0	1
17	CPV	Cabo Verde	1975	24	1	0	24
18	CUB	Cuba	1950	36	0	1	1
19	CYP	Cyprus	1960	34	1	0	54
20	DOM	Dominican Republic	1950	90	2	1	22
21	ECU	Ecuador	1950	121	1	1	33
22	ESP	Spain	1950	129	1	0	37
23	FJI	Fiji	1960	80	3	3	32
24	GEO	Georgia	1992	79	1	0	11
25	GHA	Ghana	1950	101	4	3	27
26	GMB	The Gambia	1965	49	1	1	27
27	GRC	Greece	1950	113	2	1	51
28	GTM	Guatemala	1950	105	1	0	16
29	GUY	Guyana	1960	60	1	0	22
30	HKG	Hong Kong	1960	137	1	0	23
31	HND	Honduras	1950	114	1	1	14
32	HRV	Croatia	1995	114	1	0	15
33	HUN	Hungary	1950	121	1	0	25
34	IDN	Indonesia	1950	159	2	1	18
35	IND	India	1950	125	2	1	61
36	JAM	Jamaica	1950	15	1	0	62
37	KEN	Kenya	1955	152	2	1	10
38	KGZ	Kirgistan	1992	68	1	0	4
39	KOR	South Korea	1950	130	1	0	27
40	LBN	Lebanon	1950	150	1	0	4
41	LBR	Liberia	1950	87	1	0	9
42	LBY	Libya	1960	70	1	1	1
43	LKA	Sri Lanka	1950	150	2	3	45
44	LSO	Eswatini	1960	52	2	1	15
45	MDG	Madagascar	1950	119	1	1	3
46	MDV	Maldives	1980	58	1	1	4
47	MEX	Mexico	1950	120	1	0	18
48	MKD	North Macedonia	1993	83	1	1	18
49	MLI	Mali	1960	85	2	1	21
50	MNG	Mongolia	1950	36	1	0	24

(Continued Overleaf)

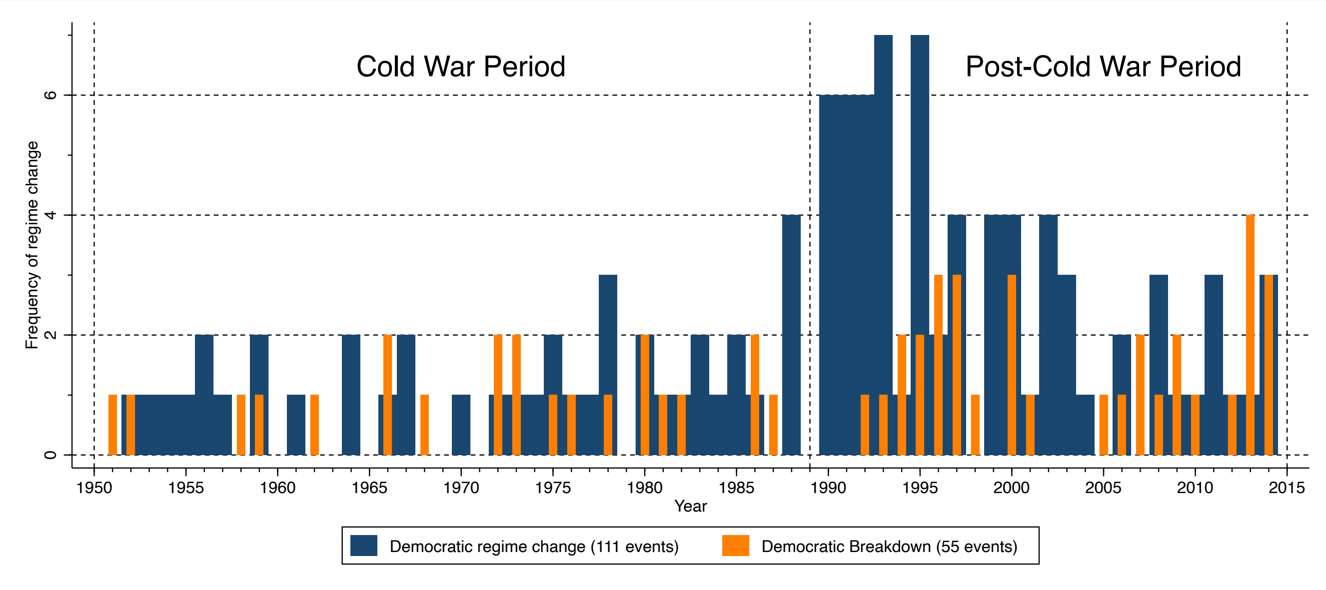
Table A-1: Sample Makeup: Exporters (continued)

	ISO	Exporter	Start	Pairs	Demo	Reversal	Years
51	MOZ	Mozambique	1975	84	1	0	20
52	MWI	Malawi	1954	88	1	0	20
53	NAM	Namibia	1980	21	1	0	25
54	NER	Niger	1960	94	3	2	16
55	NGA	Nigeria	1950	104	1	0	15
56	NIC	Nicaragua	1950	96	1	1	17
57	NPL	Nepal	1952	99	3	2	15
58	PAN	Panama	1950	83	1	0	24
59	PER	Peru	1950	130	3	2	29
60	PHL	Philippines	1950	120	1	0	27
61	PNG	Papua New Guinea	1960	32	1	0	43
62	POL	Poland	1950	127	1	0	25
63	PRT	Portugal	1950	119	1	0	39
64	PRY	Paraguay	1950	74	1	0	23
65	RUS	Russia	1992	129	1	2	2
66	SEN	Senegal	1959	69	1	0	37
67	SGP	Singapore	1950	152	1	0	15
68	SLB	Solomon Islands	1967	38	2	1	35
69	SLE	Sierra Leone	1960	79	1	0	12
70	SLV	El Salvador	1950	95	1	0	16
71	STP	Sao Tome & Principe	1975	18	1	0	16
72	SUR	Suriname	1960	46	1	1	47
73	SYC	Sychelles	1960	71	1	0	12
74	THA	Thailand	1950	162	3	3	18
75	TTO	Trinidad & Tobago	1951	44	1	0	59
76	TUN	Tunisia	1960	130	1	0	3
77	TUR	Turkey	1950	164	3	3	39
78	TWN	Taiwan	1951	146	1	0	14
79	TZA	Tanzania	1955	93	1	0	23
80	UKR	Ukraine	1992	154	1	2	10
81	URY	Uruguay	1950	95	1	1	53
82	VEN	Venezuela	1950	103	1	1	42
83	ZAF	South Africa	1950	126	1	0	20
84	ZMB	Zambia	1950	87	1	0	23
	Total			7732			

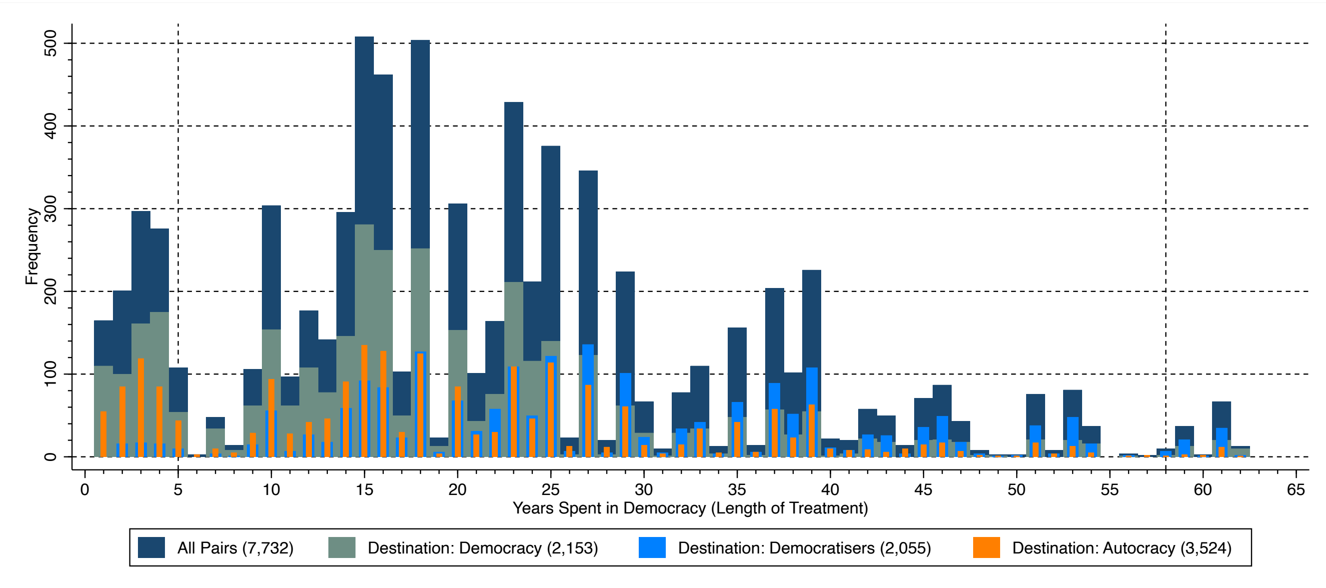
Notes: We present details of the 84 exporters which experienced regime change during 1950-2015 based on our baseline Liberal Democracy definition. 'Start' indicates the year in which the country enters the dataset, 'Pairs' the number of trade partners (with positive trade flows in at least one year), 'Demo' the number of democratic regime changes, 'Reversal' the number of autocratic reversals, and 'Years' the total number of years in democracy. We only report these statistics for countries which did not get assigned a zero weight by our M-estimator.

Figure A-1: Democratic Regime Change — Years (countries) and Length of Treatment (dyadic)

(a) Year of Regime Change (Liberal Democracy)



(b) Distribution — Years Spent in Democracy (Liberal Democracy)



Notes: In panel (a) we present the frequency of regime change by year for the 84 (exporter) countries which experienced regime change. A total of 111 democratic regime changes and 55 democratic breakdowns occurred during the 1950-2014 sample period. Panel (b) uses the dyadic data for these 84 countries (7,732 pairs) and indicates the frequency of total number of years spent in democracy on the x-axis.

B Full Results: ATET Estimates

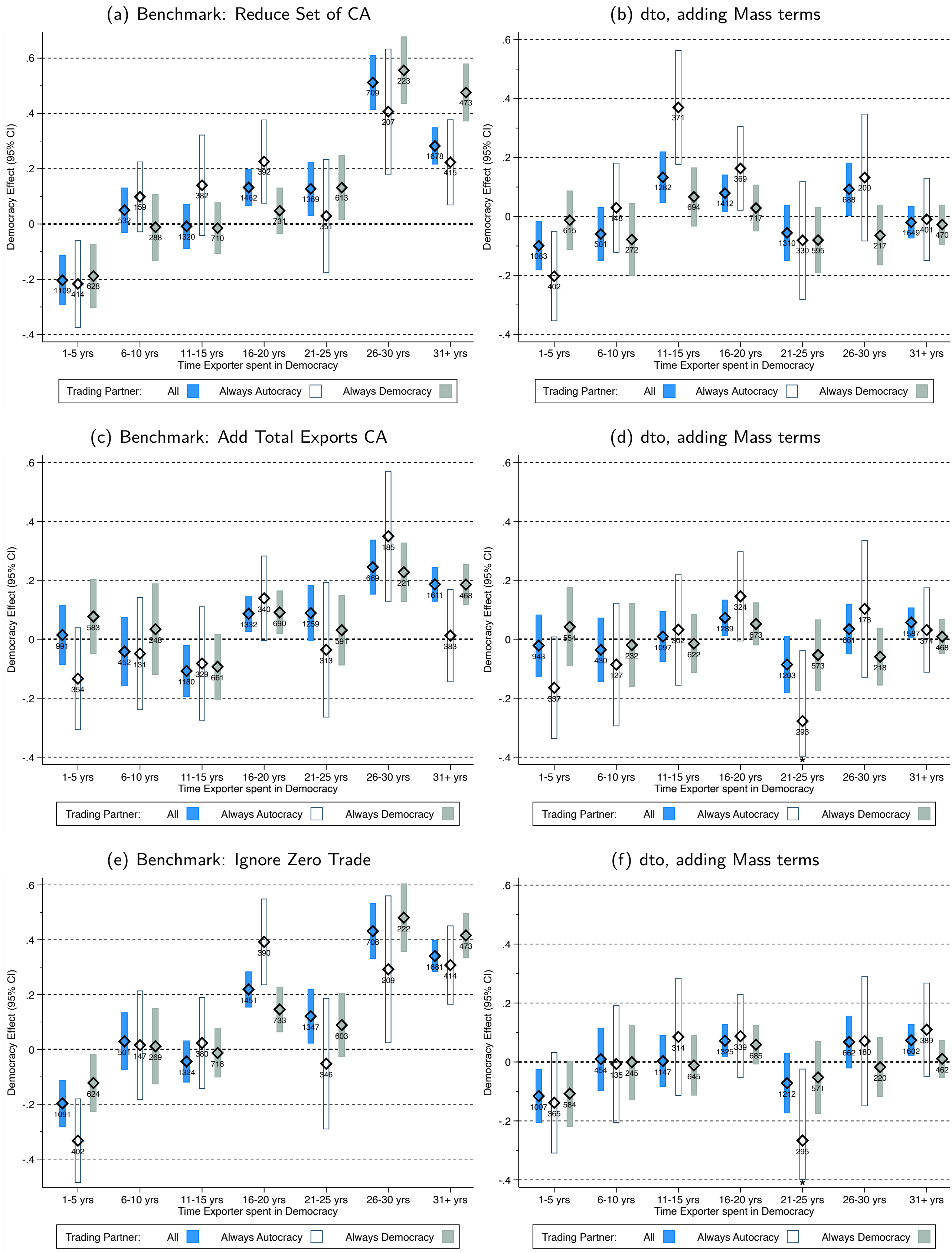
Table B-1: Average Treatment Effects and Pseudo-Alpha Test Statistics

<still to be added>

Notes: The table presents robust mean estimates (average treatment effects on the treated computed using robust regression) for all specifications and subsamples considered in the paper: Figure 1 in the main text and Figures C-1 and C-2 in the Appendix. Columns indicate the destination (in (C) and (D) exporter) regime status, including the sub-sample of 'democratisers' (Aut-Dem). The Pseudo-Alpha test is for the null hypothesis that the coefficients on the cross-section averages sum to 1, implemented using a seemingly unrelated regression framework.

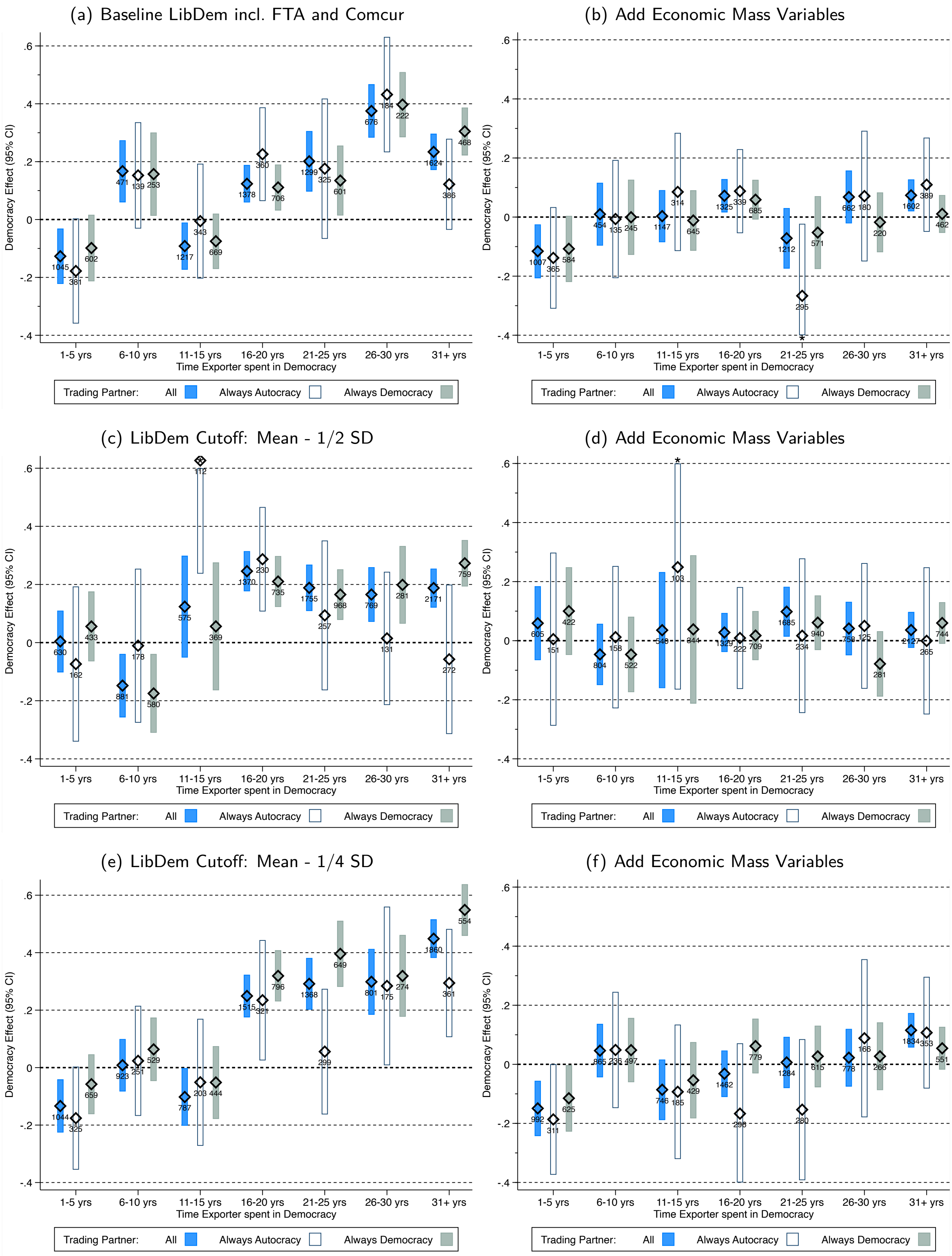
C Results: Robustness Checks

Figure C-1: Democratic Regime Change and Trade Flows — Various Specifications



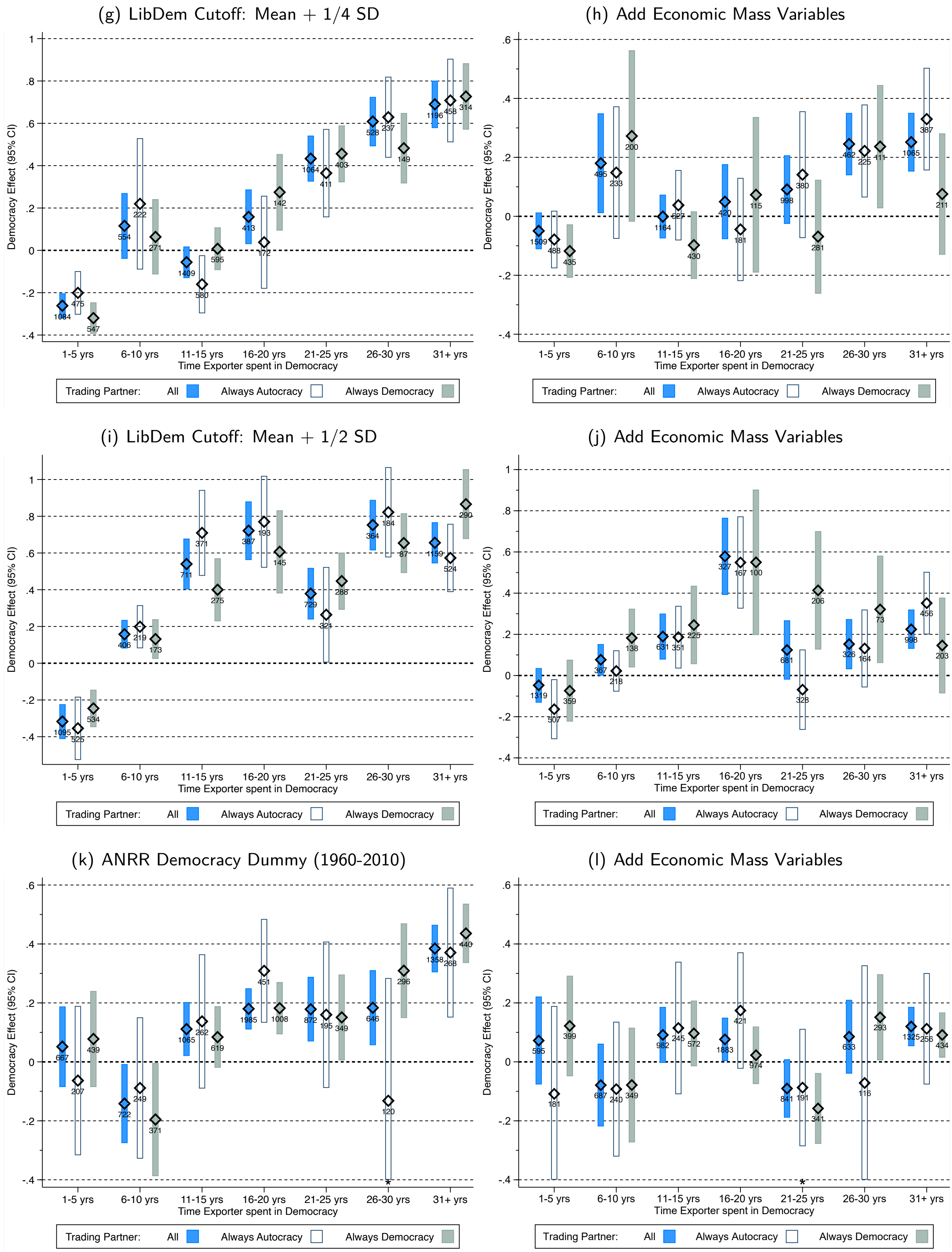
Notes: These plots present robust mean estimates for time spent in democracy (5-year averages). We distinguish trade partners which are always democracies and always autocracies. The bars are the 95% confidence intervals, and the markers are the outlier-robust means. We indicate the number of country pairs included in each estimate.

Figure C-2: Democratic Regime Change and Trade Flows — Alternative Democracy Dummies



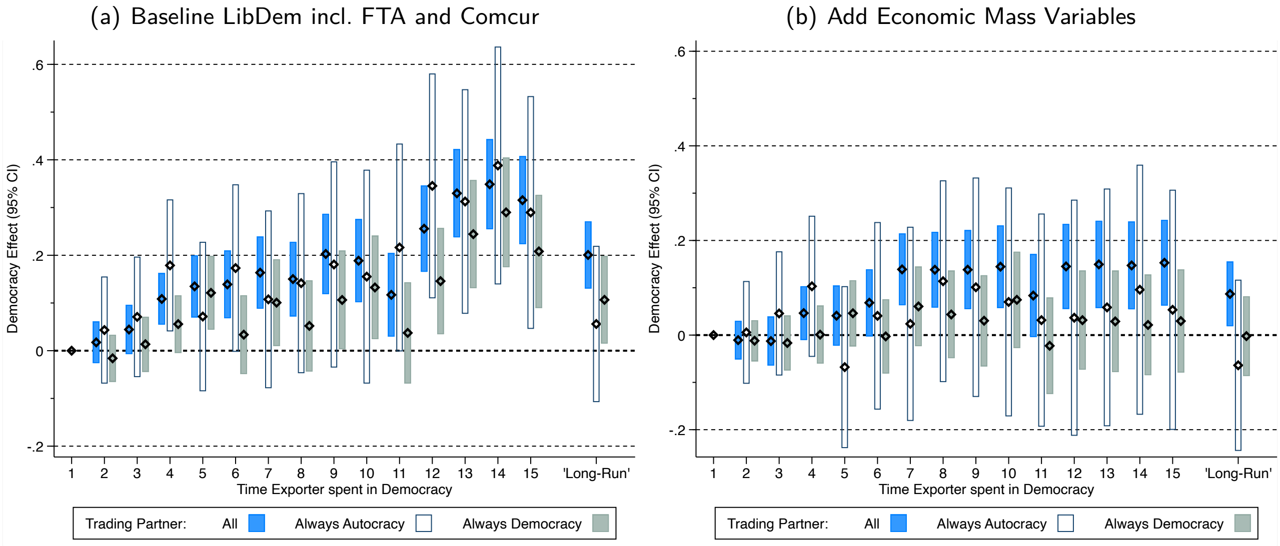
(Continued Overleaf)

Figure C-2: Democratic Regime Change and Trade Flows — Alternative Democracy Dummies (cont'd)



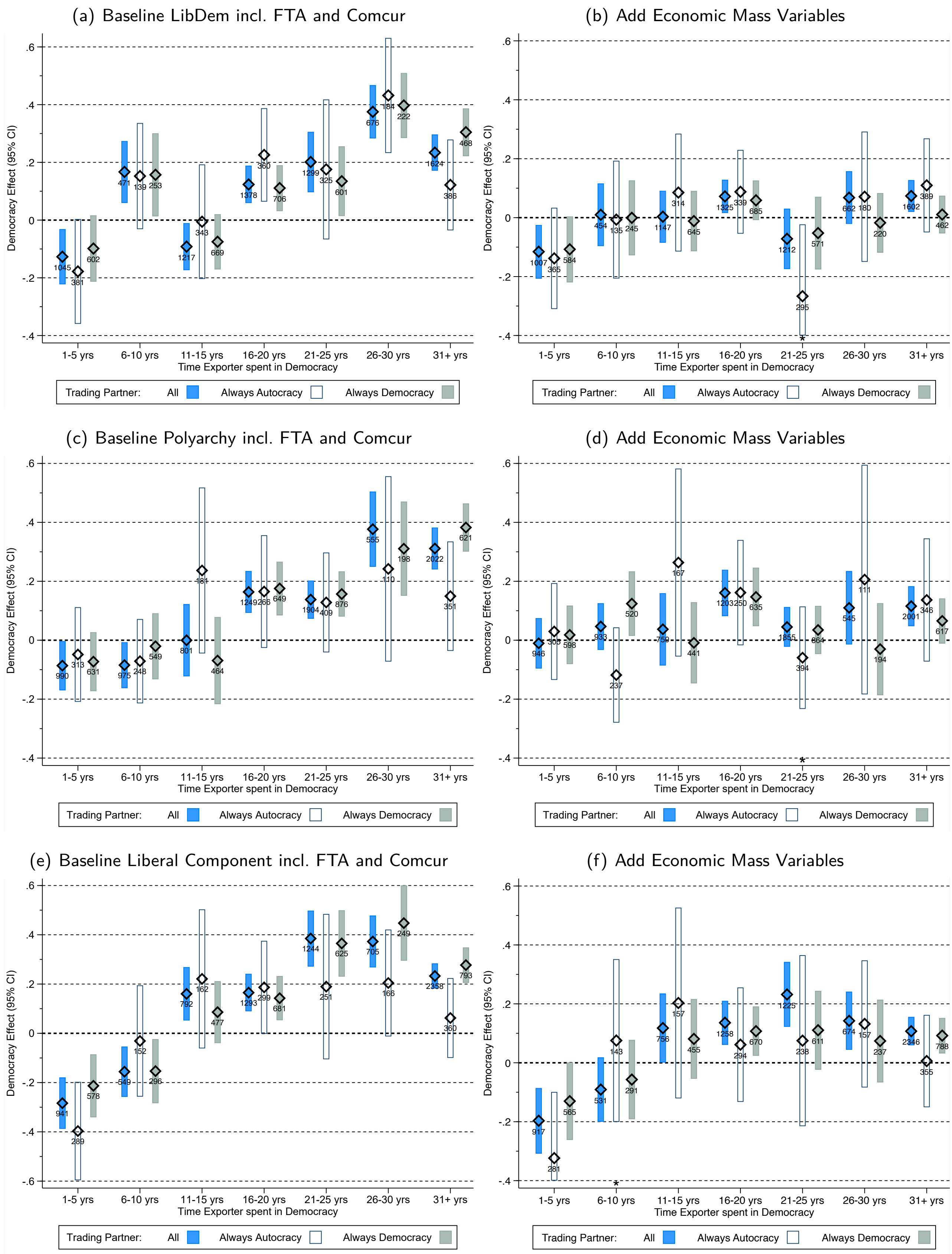
Notes: We use different definitions for our results adopting benchmark Liberal Democracy regime change (mean of the libdem index) — reprinted in panels (a) and (b) for the benchmark and model with mass variables, respectively: four versions, in panels (c) to (j) where we still use the Mean for the V-Dem LibDem index but subtract (add) 1/4 or 1/2 of a standard deviation to establish the democracy cutoff. In panels (k) and (l) we adopt the definition of Acemoglu et al (2019) which covers 1960-2010 (rather than 1950-2014). For all other details see Figure C-1.

Figure C-3: Democratic Regime Change and Trade Flows — Event Study-Style Implementation



Notes:

Figure C-4: Democratic Regime Change and Trade Flows — Building Blocks of Liberal Democracy



Notes:

D The PPML-CCE-DID Estimator

In this Appendix Section, we take more time to motivate and develop our empirical implementation to capture monadic variables in a factor-augmented heterogeneous gravity model. The first two sub-sections are a recap of the modern structural gravity empirics, before section D.3 introduces heterogeneous gravity. Our combination of the PPML-CCE and recent heterogeneous difference-in-differences estimators is presented thereafter.

D.1 The Gravity Equation

We assume a gravity relationship in the panel in which bilateral trade of exporter i to destination market j at time t is given by[30]

$$X_{ijt} = \frac{Y_{it}}{\Omega_{it}} \frac{X_{jt}}{\Phi_{jt}} \phi_{jit} \qquad \text{where} \quad 0 \leq \phi_{ijt} \leq 1. \tag{2}$$

X_{ijt} is a trade flow from an exporter to a destination market, Y_{it} is the value of production for the exporter and X_{jt} the value of expenditure in the destination market j on all source countries — the latter two are typically proxied by GDP in the exporter and destination markets, respectively.[31] ϕ_{ijt} captures the 'bilateral accessibility' for destination j and exporter i: this contains trade costs between the two markets and any variable which may affect these, including time-variant and invariant, observed and unobserved factors.

A major development in gravity modelling over the past decade following the seminal contribution by Anderson and Van Wincoop (2003) is the recognition that the conditional trade between destination j and exporter i (the conditions being the 'bilateral accessibility') cannot be viewed in isolation from the set of opportunities open to importer j in sourcing goods from exporters other than i and the relative access exporter i has to destinations other than j.[32] The multilateral resistance variables for each actor in the exchange of goods are defined in terms of the bilateral accessibility-weighted exporter capabilities and importer characteristics respectively: exporter $i = 1, \ldots, N-1$ and for importer $j = 1, \ldots, N-1$, and $i \neq j$ let

$$\Omega_{it} = \sum_{\ell=-i} \frac{\phi_{i\ell t} X_{\ell t}}{\Phi_{\ell t}} \qquad \Phi_{jt} = \sum_{\ell=-j} \frac{\phi_{\ell jt} Y_{\ell t}}{\Omega_{\ell t}} \tag{3}$$

where $-j$ and $-i$ signify that these magnitudes are not defined in reflexive terms and thus exclude destination j and exporter i from the respective MRTs of the trading relationship between these two markets.

For our derivation of the empirical gravity model, we assume a stochastic version of equation (2)

$$X_{ijt} \quad = \quad \frac{Y_{it}}{\Omega_{it}} \frac{X_{jt}}{\Phi_{jt}} \phi_{ijt} \eta_{ijt} \tag{4}$$

where η_{ijt} is an error factor with $E[\eta_{ijt} | Y_{it}, \Omega_{it}, X_{jt}, \Phi_{jt}, \phi_{ijt}] = 1$.

A very general empirical equivalent to equation (4) allows for flexible unknown parameters on the observable mass and accessibility variables:

$$\begin{aligned} X_{ijt} \quad = \quad & exp[\beta^i_{ijt} \, ln(Y)_{it} + \beta^n_{ijt} \, ln(X)_{jt} + \gamma_{ijt} \, ln(\phi)_{ijt} \\ & + ln(\Omega)_{it} + ln(\Phi)_{jt}] \, \eta_{ijt}, \end{aligned} \tag{5}$$

where superscripts are used to identify the coefficient of exporter versus importer GDP/expenditure. This

[30] This exposition builds on the discussion in Santos Silva and Tenreyro (2006), and Yotov et al. (2016) for the gravity model, and Cameron and Trivedi (1998) for Poisson regression.

[31] If X_{ij} is merchandise trade then theory-consistency dictates Y_i to be gross production of traded goods (not simply value-added/GDP) and X_j the apparent consumption of goods, production plus imports minus exports (Head and Mayer, 2014).

[32] This network of dependencies is formalised by econometricians as the deviation from the assumption of 'cross-section weak dependence' (Andrews, 2005; Chudik, Pesaran, and Tosetti, 2011; Chudik and Pesaran, 2015b).

specification is the most general empirical model possible where all unknown parameters on observable variables (β^i, β^j, γ) vary at the pair level and over time. We demonstrate how this relates to the models employed in the existing literature by adopting parameter restrictions.

D.2 Model Restrictions and Pooled Estimation

A first restriction, adopted in the vast majority of studies, is to assume the gravity model estimates are fixed *across time*:

$$\begin{aligned} X_{ijt} \;&=\; exp[\beta^i_{ij}\, ln(Y)_{it} + \beta^n_{ij}\, ln(X)_{jt} + \gamma_{ij}\, ln(\phi)_{ijt} \\ &\quad + ln(\Omega)_{it} + ln(\Phi)_{jt}]\, \eta_{ijt}, \end{aligned} \tag{6}$$

A notable exception is the study by Klasing, Milionis, and Zymek (2015) who allow for time-variation in three distinct regimes over their long panel from 1870 to 2005. In our analysis of the post-WWII period we first follow the bulk of the literature and estimate policy effects which are specified as time-invariant; later on we *partly* relax the assumption of fixed parameters over time by adopting 20-year rolling regression windows.

Conventionally, further restrictions in the panel gravity literature are to assume common parameters on the observable variables ($\beta^i_{ij} = \beta^i$, $\beta^n_{ij} = \beta^n$, $\gamma_{ij} = \gamma$). Pairwise fixed effects (δ_{ij}) are added to capture trade policy endogeneity (Baier and Bergstrand, 2007). Additionally including exporter-time (ω_{it}) and importer-time (ψ_{jt}) fixed effects can capture the MRTs (Hummels, 2001; Anderson and van Wincoop, 2003; Feenstra, 2004) — with implications for β^i and β^j (see below).

$$\begin{aligned} X_{ijt} \;&=\; exp[\beta^i\, ln(Y)_{it} + \beta^j\, ln(X)_{jt} + \gamma\, ln(\phi)_{ijt} + \delta_{ij} \\ &\quad + \omega_{it} + \psi_{jt}]\, \eta_{ijt}. \end{aligned} \tag{7}$$

Thus δ_{ij}, ϕ_{jt} and ω_{it} are the unknown parameters estimated on the various fixed effects in the reduced-form panel gravity model.

D.3 Heterogeneous Parameter Estimation

A practical difficulty arises if a more flexible specification for the observable variables such as that laid out in equation (5) on the one hand (β^i_{ij}, β^n_{ij}, γ_{ij}), and the aforementioned recommended practice to capture MRTs on the other are to be combined: for pairwise heterogeneity in the economic mass and policy variable parameters it is most convenient to estimate equation (5) *separately* for each country pair. However, we cannot include fixed effects for *all exporters and all importers* in an equation for a single importer-exporter pair, let alone interacted with time dummies. Existing studies in the literature which allow for trade policy heterogeneity employ interaction effects (e.g. Baier, Bergstrand and Clance, 2018) or maintain a set of fixed effects in a PPML model but use pair-specific trade policy dummies (Baier, Yotov and Zylkin, 2019). An alternative approach is to draw on recent insights from the panel time series literature (e.g. Pesaran, 2006; Bai, 2009) and to employ a multi-factor error structure to capture the unobserved MRTs.

First, we bring the error factor on the inside of the exponential function to capture all unobservables u_{ijt}:

$$\begin{aligned} u_{ijt} \;&=\; \delta_{ij} + \omega_{it} + \psi_{jt} + ln(\eta)_{ijt} \\ &\equiv\; \delta_{ij} + \omega_{it} + \psi_{jt} + \varepsilon_{ijt}. \end{aligned} \tag{8}$$

Next, the dimensionality problem of dealing with a large number of unknown parameters (the number of ω_{it} and ϕ_{jt} swiftly add up to thousands of directional-time dummies) in a country-pair equation with about 70 post-WWII time series observations can be solved by imposing more structure on these unobservables. In the macro panel econometric literature it is widely acknowledged that a small number of common factors with heterogeneous factor loadings can represent large datasets of macro variables. For instance, in the forecasting literature Stock and Watson (2002) have shown that 149 macroeconomic time series can be reduced to two or three principal components. Furthermore, Bai (2009) discusses several macro-, microeconometric and finance applications where common factors can be employed to model unobserved time-varying heterogeneity in a tractable way.[33]

In the case at hand, we posit that the economic mass and accessibility variables along with the trade flows and MRTs are all driven by a small number of common factors with heterogeneous factor loadings across country pairs. Thus, we argue that a small number of unobserved common factors $\boldsymbol{f}_t$, each with country pair-specific factor loadings $\boldsymbol{\varphi}_{ij}$, can account for the evolution of trade flows, GDP, etc. In the notation introduced above:

$$
\begin{aligned}
u_{ijt} &= \delta_{ij} + \omega_{it} + \psi_{jt} + \varepsilon_{ijt} &(9)\\
&\approx \delta_{ij} + \boldsymbol{\varphi}'_{ij}\boldsymbol{f}_t + \varepsilon_{ijt}, &(10)
\end{aligned}
$$

where an 'approximate factor structure' is represented by a set of common factors $\boldsymbol{f}$, and the associated factor loadings $\boldsymbol{\varphi}$.

The insight gained in the recent panel time series literature from this setup in the *linear* regression case is that the unobserved common factors can be captured by observables, either via principal component analysis (Bai, 2009) or using cross-section averages of the dependent and independent variables (Pesaran, 2006). We briefly develop the latter approach (the 'common correlated effects' or CCE estimator) and provide a mathematical indication of the intuition at play — this is for ease of illustration since it will only be a small step in terms of implementation from the linear model to a generalised linear one (Boneva and Linton, 2017).

For simplicity we assume a double-index of t for the time series and i for the cross-section — we can think of the latter as a placeholder for the country pair as the unit of analysis like in the gravity model, i.e. $i \equiv ij$. Let

$$
\begin{aligned}
y_{it} &= \alpha_i + \beta_i x_{it} + u_{it} \qquad u_{it} = \lambda_i f_t + \varepsilon_{it} &(11)\\
x_{it} &= \delta_i + \phi_i f_t + \varrho_i g_t + e_{it}, &(12)
\end{aligned}
$$

where ε and e are white noise processes. This setup indicates that the regressor x is driven by the same common factor f as the dependent variable y, albeit with different parameters.[34] In addition there are some factors $\boldsymbol{g}$ which only drive x but not y. This setup is standard in the macro panel literature and we refer to the studies in footnote 33 for details on factor evolution, parameter distributions, etc. It is clear from equations (11) and (12) that x is endogenous and that failing to account for the presence of the unobserved common factors will lead to omitted variable bias.[35]

Pesaran's (2006) approach posits that the unobserved common factor f can be captured by the cross-

[33] The multifactor error structure has been applied to capture country-specific time-varying total factor productivity (Eberhardt and Presbitero, 2015), time-varying absorptive capacity (De Visscher et al, 2020), or knowledge spillovers in the analysis of sector-level production functions augmented with sectoral R&D stock (Eberhardt, Helmers, and Strauss, 2013).

[34] In a setup with multiple factors we can instead assume that only a subset of $\boldsymbol{f}$ 'overlaps' between the two equations.

[35] Solving the x equation for f and plugging this into the y equation yields

$$
\begin{aligned}
y_{it} &= \alpha_i + \beta_i x_{it} + \lambda_i \phi_i^{-1}(x_{it} - \delta_i - \psi_i g_t - e_{it}) + \varepsilon_{it} &(13)\\
&= \alpha_i - \lambda_i \phi^{-1}\delta_i + (\beta_i + \lambda_i \phi_i^{-1})\, x_{it} - \lambda_i \phi_i^{-1}\psi_i g_t - \lambda_i \phi_i^{-1} e_{it} + \varepsilon_{it} \\
&= \eta_i + \theta_i x_{it} + \nu_{it}, &(14)
\end{aligned}
$$

where in the final line we reparameterise. Crucially, unless $\lambda_i \phi_i^{-1} = 0$ we can see that β_i is unidentified. The asymptotic bias will be a function of the (relative) 'strength' of the factors in their impact on y and x in panel member i.

section averages of y and x provided the cross-section dimension of the panel is not too small.[36] A simple algebraic derivation can provide intuition for the mechanism at work: take the cross-section average of equation (11) and solve it for the common factor f:

$$\bar{y}_t \quad = \quad \bar{\alpha} + \bar{\beta}\bar{x}_t + \bar{\lambda} f_t \quad \Leftrightarrow \quad f_t = \bar{\lambda}^{-1}(\bar{y}_t - \bar{\alpha} - \bar{\beta}\bar{x}_t), \tag{15}$$

where the bars indicate cross-section averages and the error term disappears since $\bar{\varepsilon} = 0$ by assumption. Next, plug the expression for f back into the original equation (11)

$$\begin{aligned} y_{it} &= \alpha_i + \beta_i x_{it} + \lambda_i \bar{\lambda}^{-1}(\bar{y}_t - \bar{\alpha} - \bar{\beta}\bar{x}_t) + \varepsilon_{it} & (16)\\ &= \alpha_i - \lambda_i \bar{\lambda}^{-1}\bar{\alpha} + \beta_i x_{it} + \lambda_i \bar{\lambda}^{-1}\bar{y}_t - \lambda_i \bar{\lambda}^{-1}\bar{\beta}\bar{x}_t + \varepsilon_{it}, & \\ y_{it} &= \varpi_i + \beta_i x_{it} + \zeta_i \bar{y}_t + \vartheta_i \bar{x}_t + \varepsilon_{it}, & (17) \end{aligned}$$

where we reparameterize in the last line. Thus the unobserved common factor f can be captured by the cross-section averages of y and x, while the heterogeneous impact of f across i can be captured by estimating equation (17) separately for each panel member — the principle extends to multiple factors. A Mean Group estimator following Pesaran and Smith (1995) captures the central tendency of the panel and provides a convenient comparison with alternative pooled empirical models:

$$\hat{\beta}^{MG} = N^{-1} \sum_{i=1}^{N} \hat{\beta}_i. \tag{18}$$

Inference for the Mean Group estimates is based on a simple nonparametric variance estimator (Pesaran and Smith, 1995; Pesaran, 2006):[37]

$$\widehat{\Omega}^{MG} = \frac{1}{N-1} \sum_{i=1}^{N} (\hat{\beta}_i - \hat{\beta}^{MG})(\hat{\beta}_i - \hat{\beta}^{MG})'. \tag{19}$$

Boneva and Linton (2017) extend the above setup from the linear model to a setting where the outcome variable is discrete. They show that the CCE approach can be applied to a probit model by including only the cross-section averages of the observed regressors: under the assumption that the unobserved factors are contained in the span of the cross-section averages of the regressors they derive asymptotic results for the large T, large N case as well as the consistency and asymptotic normality of the Mean Group estimator of the individual-specific estimates.

The same principle can be applied to a generalised linear model: in our case, we begin by assuming the exponential mean function incorporating a multi-factor error structure[38]

$$E[y_{it}|x_{it}, \boldsymbol{f}_t] = \mu_{it} = exp[\alpha_i + \beta_i x_{it} + \boldsymbol{\lambda}_i' \boldsymbol{f}_t]. \tag{20}$$

The Poisson Maximum Likelihood (PML) estimator assumes that the distribution of y given x (and the factors) is Poisson (i.e. a count variable), but it is widely recognised that the data generating process is not required to be Poisson for this estimator to be consistent (Cameron and Trivedi, 1998; Santos Silva and Tenreyro, 2006), in which case it is referred to as a Poisson Pseudo Maximum Likelihood (PPML) estimator. Our implementation uses the estimator in the single time series, namely of the pair-wise gravity equation with exponential mean

[36] Additional robustness of this approach to nonstationary factors, structural breaks, and additional spatial dependence, among other aspects, is discussed in Kapetanios, Pesaran, and Yamagata (2011), Chudik, Pesaran and Tosetti (2011), Chudik and Pesaran (2015a) and Westerlund (2019).

[37] In practice we follow the standard in the literature and employ robust means (Hamilton, 1992) to reduce the effect of outliers.

[38] We thank Lena Boneva and Oliver Linton for sharing the rough derivations for a CCE-augmentation in this more general case.

function, where the common factors are replaced by cross-section averages of the regressors, which we refer to as PPML-CCE:

$$\begin{aligned}\mu_{ijt} &= exp[\alpha_{ij} + \beta^{i}_{ij}\, ln(Y)_{it} + \beta^{n}_{ij}\, ln(X)_{jt} + \gamma_{ij}\, ln(\phi)_{ijt} \\ &\quad +\delta_{ij}\, \overline{ln(Y)}_t + \kappa_{ij}\, \overline{ln(\phi)}_t] + \epsilon_{ijt} \qquad \forall\, ij,\, j \neq i.\end{aligned} \tag{21}$$

In our case, the accessibility term (any variable that may affect bilateral trade costs) is replaced by dummies for democratic regime change in i and j respectively along with dyadic dummies for free trade agreements and currency unions (in robustness checks we omit the latter two since they may be viewed as 'bad controls').[39] In practice, where exporter and destination mass are proxied using their respective GDP, the cross-section averages are identical, $\overline{ln(Y)}_t = \overline{ln(X)}_t$. In our difference-in-differences implementation below these come from distinct control samples and are thus separately identified.

Our empirical estimates below are thus based on cross-section average-augmented time series PPML regressions at the country-pair level which are subsequently averaged following the Mean Group principle; inferential statistics for the PPML-CCE Mean Group estimate are computed using the variance estimator in (19): Boneva and Linton (2017) have shown the simple nonparametric variance estimator still applies in the generalised linear model setup.

Estimation of the PPML-CCE model can be seen as an improvement on current practice for two reasons. First, the presence of common factors, with heterogeneous loadings across country pairs, offers a flexible way to account not only for the MRTs, but also so-called globalisation effects, and other forms of spatial dependence. Second, using common factors as proxies for spatial dependence allows the estimation of country-specific (monadic) variables — thus in addition to dyadic determinants such as those contained in $ln(\phi)_{ijt}$, we can also introduce country-specific variables of interest.

D.4 A Difference-in-Differences Gravity Model

Our exposition so far has focused on heterogeneous gravity without giving too much thought to the empirical practices in the democracy-growth literature, where economists favour using *binary* indicators for democracy versus autocracy (Papaioannou and Siourounis, 2008; Acemoglu et al, 2019; Eberhardt, 2022; Boese-Schlosser and Eberhardt, 2023, 2024). We now introduce a combination of the PPML-CCE model with a factor-augmented difference-in-differences estimator following Chan and Kwok (2022): let D_{it} be a dummy for democratic regime change equal to 0 when country i is in autocracy and 1 if they are in democracy — we refer to the shift from 0 to 1 as 'treatment'. The Chan and Kwok (2022) principal component DID (PCDID) estimator estimates the causal effect of a treatment in country-specific regressions of treated countries only. While this appears to capture only the first of the difference-in-differences, the comparison with the control sample (in our case those countries which never experienced democratic regime change) is achieved by the inclusion of unobserved common factor proxies extracted from the control sample of never-democratisers. Crucially, this heterogeneous difference-in-differences estimator does not require the 'parallel trends' assumption to hold, but relies on a much weaker assumption of 'weak parallel trends' between treated and control samples — we provide more details in the next section where we develop a suitable test in the following subsection.

In the context of democratic regime change and the PPML-CCE gravity model of trade flows, we make some adjustments to the way the Chan and Kwok (2022) PCDID is implemented: first, we do not *estimate* the unobserved common factors from auxiliary regressions in the control sample of never-democratisers. Extracting several principal components via PCA from the residuals of a PPML regression is conceptually difficult (given

[39] We know that the dependent variable, conditional on the regressors, can be heteroskedastic, and serially correlated and that the PPML estimator maintains its consistency in the presence of serial correlation if the exponential conditional mean is correctly specified (Cameron and Trivedi, 1998: 226) — the latter is the requirement for any PML or PPML estimator.

that these are not linear models) and in practice raises concerns over how many estimated factors to include. Instead, we use cross-section averages[40] of GDP and population in line with Boneva and Linton (2017). Second, these cross-section averages are constructed from the respective control samples of (i) exporters and (ii) destinations: since the trade flow panel is not symmetric (i may be exporting to j but j does not export to i) these two cross-section averages are not identical.

Our PPML-CCE-DID estimator is then implemented as follows:

$$\begin{aligned}\mu_{ijt} \quad = \quad & exp[\alpha_{ij} + \gamma^1_{ij}\,\text{FTA}_{ijt} + \gamma^2_{ij}\,\text{Common Currency}_{ijt} + \theta_{ij} D_{it} + \eta_{ij} D_{jt} \\ & +\delta^i_{ij}\,\overline{ln(Y)}^i_t + \kappa^i_{ij}\,\overline{ln(Pop)}^i_t + \delta^j_{ij}\,\overline{ln(Y)}^j_t + \kappa^j_{ij}\,\overline{ln(Pop)}^j_t + \epsilon_{ijt}] \qquad \forall\, ij,\, j \neq i.\end{aligned} \tag{22}$$

We have pair fixed effect (α_{ij}) and the familiar dyadic variables for FTA and common currency union. The two democracy dummies (D_{it} for exporter i and D_{jt} for destination j) are accompanied by two sets of cross-section averages, which are constructed from the exporter i and destination j control samples of countries which never democratised, respectively. We do not include cross-section averages of the indicator variables in the first line of equation (22) since this practice is likely to induce severe bias in the estimation equation (Juodis et al, 2021). Our coefficient of interest is θ_{ij}, which is estimated (separately) in each trade flow equation of country i with each of its trade partners j — as is standard in the heterogeneous DID literature, we estimate equation (22) only for equations where either country i or country j experienced regime change.

We cannot estimate the democracy effect in country-pairs where trade flows remained zero throughout the sample period, whether they experienced regime change or not: this is necessitated by our heterogenous regression implementation. We do however include the GDP and population variables for these country pairs in the computation of the cross-section averages. More worryingly, excluding country pairs where the exporter or importer did experience regime change but nevertheless, trade between the pair never materialised, will likely induce an upward (selection) bias in our results. We therefore adjust the model to include cross-section averages from those country pairs where the exporter or importer country did experience regime change but where trade flows remained zero. Our final empirical implementation is thus:

$$\begin{aligned}\mu_{ijt} \quad = \quad & exp[\alpha_{ij} + \gamma^1_{ij}\,\text{FTA}_{ijt} + \gamma^2_{ij}\,\text{Common Currency}_{ijt} + \theta_{ij} D_{it} + \eta_{ij} D_{jt} \\ & +\delta^i_{ij}\,\overline{ln(Y)}^i_t + \kappa^i_{ij}\,\overline{ln(Pop)}^i_t + \delta^j_{ij}\,\overline{ln(Y)}^j_t + \kappa^j_{ij}\,\overline{ln(Pop)}^j_t \\ & +\delta^{i0}_{ij}\,\overline{ln(Y)}^{i0}_t + \kappa^{i0}_{ij}\,\overline{ln(Pop)}^{i0}_t + \delta^{j0}_{ij}\,\overline{ln(Y)}^{j0}_t + \kappa^{j0}_{ij}\,\overline{ln(Pop)}^{j0}_t + \epsilon_{ijt}] \qquad \forall\, ij,\, j \neq i,\end{aligned} \tag{23}$$

where the superscript $i0$ ($j0$) indicates the cross-section averages or associated parameters from exporter (destination) countries which did experience regime change but where trade flows between i and j remained zero throughout the sample period. This specification is even more demanding on the data, with 13 parameters to be estimated in each country-pair regression.

[40] Eberhardt (2022) adopts the same strategy in a standard (monadic) panel dataset to study democracy and growth.